

Topological Energy Condensate Theory (TECT): 6

Jusang Lee^{1,*}

¹*Independent Researcher, Incheon, Korea*
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I. WHY THE MINIMAL MICROSCOPIC TECT MODEL WAS CHOSEN

A. Statement of the problem

Before continuing with further consequences, we must justify the microscopic model choice itself. The correct question is not:

“Which equation looks plausible?”

but rather:

“Which minimal microscopic equation can simultaneously support all already-established TECT requirements?”

At the current stage of the program, the following structures are already fixed at the theorem level:

1. a finite- q shell instability is required for shell condensation and BCC-type pattern formation;
2. the internal geometry must reduce to $\mathbb{CP}^2$;
3. the shell-level Dirac mechanism requires an enriched internal doublet and gradient-linear microscopic tensor couplings;
4. the low-energy limit should admit a relativistic-type wave/elastic sector.

Therefore the microscopic model must be selected by *constraint satisfaction*, not by aesthetic preference.

B. Constraint I: finite- q shell instability

TECT requires a condensed background built from a nonzero wavevector shell. Thus the microscopic linear dispersion must have a minimum at

$$|k| = q_* > 0.$$

A purely Klein–Gordon-type quadratic spatial operator

$$\partial_t^2 \Psi = -m^2 \Psi + c^2 \nabla^2 \Psi$$

cannot do this, because its dispersion

$$\omega^2(k) = m^2 + c^2 |k|^2$$

has its minimum at $k = 0$.

Therefore a finite-shell model necessarily requires a quartic spatial derivative term. The lowest-order rotationally invariant possibility is

$$Z \nabla^2 \Psi - Y \nabla^4 \Psi.$$

* jusanglee@kakao.com

The corresponding linear dispersion is

$$\omega^2(k) = \mu^2 + Z|k|^2 + Y|k|^4. \quad (1)$$

If

$$Z < 0, \quad Y > 0, \quad (2)$$

then $\omega^2(k)$ has a nonzero shell minimum at

$$q_*^2 = -\frac{Z}{2Y}. \quad (3)$$

Hence the finite-shell requirement by itself already forces the microscopic equation into the Brazovskii class.

C. Constraint II: quartic stabilization

A shell instability alone is not enough. The condensed phase must be bounded and stable.

For a local microscopic model with a $U(3)$ -invariant amplitude sector, the lowest stabilizing nonlinearity is

$$(\Psi^\dagger \Psi)^2.$$

At the equation-of-motion level this produces the cubic field term

$$-g(\Psi^\dagger \Psi)\Psi, \quad g > 0.$$

Thus the minimal stable nonlinear completion is

$$\partial_t^2 \Psi = -\mu^2 \Psi + Z \nabla^2 \Psi - Y \nabla^4 \Psi - g(\Psi^\dagger \Psi)\Psi.$$

Any lower-order local nonlinearity either violates the assumed internal symmetry or does not generically stabilize the amplitude.

D. Constraint III: internal $\mathbb{CP}^2$ geometry

TECT does not merely require shell condensation. It also requires an internal projective geometry whose defect sector leads to the already-established $\mathbb{CP}^2$ structure.

Let $\Psi \neq 0$. Write

$$\Psi(x, t) = \rho(x, t) z(x, t), \quad \rho \geq 0, \quad z^\dagger z = 1. \quad (4)$$

Then z is defined only up to a phase,

$$z \sim e^{i\chi} z.$$

Therefore the internal direction lives on

$$[z] \in \mathbb{CP}^2. \quad (5)$$

This is the decisive reason for choosing a complex triplet field: a single complex scalar would give only S^1 phase; a complex doublet would give $\mathbb{CP}^1 \simeq S^2$; the minimal field producing $\mathbb{CP}^2$ is precisely

$$\Psi \in \mathbb{C}^3.$$

Thus the complex triplet is not optional. It is the minimal microscopic carrier of the required TECT internal geometry.

E. Constraint IV: enriched shell-Dirac mechanism

The shell-level Dirac mechanism already proved in the previous stages does not arise from a plain scalar shell theory alone. What was proved is that the decisive microscopic structure is a gradient-linear enriched term of the form

$$\delta\mathcal{L}_{\text{enr}} = \mathbf{N}_{ij}^a (\partial_i \phi_0) (-i\partial_j) \otimes s^a, \quad (6)$$

with an internal Pauli sector s^a , $a = 1, 2, 3$, coming from a complex internal doublet.

Therefore the microscopic theory must support, after condensation and reduction, at least:

1. a nontrivial internal projective sector;
2. a reduced complex doublet inside that sector;
3. gradient-linear couplings between the shell background and the internal doublet.

A plain real scalar Brazovskii field cannot provide this. A complex triplet field can, because after amplitude-direction splitting it naturally gives a projective orientation field $z \in \mathbb{CP}^2$, whose low-energy internal subspace may reduce to the required shell-relevant complex doublet.

So the complex triplet is the minimal field content compatible with the already-closed shell-Dirac framework.

F. Constraint V: second-order time dynamics

The microscopic TECT equation should serve as a true condensate field equation, not merely as a static free-energy minimization condition. Moreover, the long-wavelength sector should admit a relativistic-type elastic/wave limit.

Therefore the most natural minimal time structure is second order:

$$\partial_t^2 \Psi.$$

A first-order-in-time Schrödinger/Gross–Pitaevskii form is possible as an effective nonrelativistic reduction, but it is less natural as the fundamental pre-condensation microscopic law if one wants an emergent Lorentz-type sector in the infrared.

Hence the minimal microscopic TECT equation should be of wave type rather than Schrödinger type.

G. Reconstruction from the constraints

Combining the previous constraints, the field must be:

1. complex triplet-valued:

$$\Psi \in \mathbb{C}^3;$$

2. local and rotationally invariant before condensation;
3. second order in time;
4. fourth order in space to support a nonzero shell;
5. quartically stabilizing in the amplitude.

Under these assumptions, the unique lowest-order local equation is

$$\partial_t^2 \Psi = -\mu^2 \Psi + Z \nabla^2 \Psi - Y \nabla^4 \Psi - g(\Psi^\dagger \Psi) \Psi. \quad (7)$$

This is the complex-triplet Brazovskii wave equation.

H. Why lower-complexity alternatives fail

It is useful to state explicitly why simpler choices are rejected.

- a. Real scalar field.* A real scalar field may support a finite- q shell, but it does not carry an internal projective manifold $\mathbb{CP}^2$, nor the enriched internal doublet needed for the shell-Dirac mechanism.
- b. Complex scalar field.* A complex scalar field gives only a $U(1)$ phase degree of freedom and cannot produce $\mathbb{CP}^2$.
- c. Complex doublet.* A complex doublet produces $\mathbb{CP}^1$, not $\mathbb{CP}^2$, so it is too small for the internal geometry already singled out in TECT.
- d. Second-order spatial model only.* A model with only ∇^2 cannot produce a finite shell and therefore cannot underlie BCC shell condensation.

Thus each simpler alternative fails at least one already-established TECT requirement.

I. Why higher-complexity alternatives are not minimal

One could of course add:

- higher spatial derivatives;
- anisotropic derivative terms;
- nonlocal kernels;
- higher-order self-interactions;
- explicit symmetry-breaking internal couplings.

But none of these is forced by the already-established TECT requirements at the present stage. Therefore they belong to possible UV refinements, not to the minimal microscopic core.

This is exactly why (7) is the correct current starting point: it is the smallest local equation satisfying all presently known constraints.

J. Physical interpretation of the model choice

The chosen microscopic equation should be read as the simultaneous encoding of four structural facts:

1. $\partial_t^2 \Psi$: genuine condensate dynamics;
2. $Z\nabla^2 - Y\nabla^4$: finite-shell pattern formation;
3. $-g(\Psi^\dagger \Psi)\Psi$: amplitude saturation and condensate stability;
4. $\Psi \in \mathbb{C}^3$: internal projective geometry $\mathbb{CP}^2$.

In this sense, the model is not an arbitrary ansatz. It is the minimal common solution of the currently closed TECT constraints.

K. Honest status

What is justified here is not absolute UV uniqueness. It is uniqueness within the minimal local truncation compatible with the theorem-level TECT data.

So the precise conclusion is:

$$\boxed{\text{Within the minimal local, isotropic, complex-triplet truncation,}} \tag{8}$$

$$\boxed{\text{the microscopic TECT candidate is essentially forced to be the complex-triplet Brazovskii wave equation.}} \tag{9}$$

This is the correct reason for choosing the model before proceeding to detailed consequences.

II. FINAL SUMMARY

The microscopic TECT model was chosen by enforcing all already-established structural requirements simultaneously.

A viable microscopic equation must:

1. generate a finite- q shell instability, hence it must contain

$$Z\nabla^2 - Y\nabla^4$$

with

$$Z < 0, \quad Y > 0;$$

2. stabilize the condensate amplitude, hence it must contain the quartic self-interaction

$$-g(\Psi^\dagger\Psi)\Psi, \quad g > 0;$$

3. produce the internal projective geometry

$$\mathbb{CP}^2,$$

which minimally requires a complex triplet field

$$\Psi \in \mathbb{C}^3;$$

4. provide a genuine microscopic condensate dynamics compatible with a relativistic-type infrared sector, hence it is naturally second order in time:

$$\partial_t^2\Psi.$$

Therefore, within the minimal local isotropic truncation, the microscopic TECT candidate is essentially forced to be

$$\partial_t^2\Psi = -\mu^2\Psi + Z\nabla^2\Psi - Y\nabla^4\Psi - g(\Psi^\dagger\Psi)\Psi.$$

This model is chosen not because it is merely convenient, but because it is the minimal equation that simultaneously supports shell condensation, BCC-type pattern formation, condensate stability, and the internal $\mathbb{CP}^2$ geometry required by the shell-enriched TECT fermion program.

III. MICROSCOPIC RECONSTRUCTION OF THE MINIMAL TECT EQUATION

A. Goal

The purpose of this section is to stop circling around the shell-reduced consequences and instead reconstruct the actual minimal microscopic TECT dynamics.

We seek a fundamental field equation for a microscopic condensate field

$$\Psi(x, t)$$

such that the following already-established TECT requirements are built in from the start:

1. translation invariance;
2. pre-condensation spatial isotropy;
3. finite-shell instability at a nonzero wave number q_* ;
4. quartic stabilization of the condensate;
5. an internal sector reducing to $\mathbb{CP}^2$;
6. a relativistic-type second-order time dynamics.

We do not claim here that the full UV completion is uniquely derived with no assumptions. What we do prove is that under the minimal locality-and-symmetry truncation, the microscopic TECT equation is essentially forced into a unique lowest-order form.

B. Microscopic field content

We take the microscopic condensate field to be a complex triplet

$$\Psi(x, t) \in \mathbb{C}^3. \quad (10)$$

The reason for this choice is structural:

1. the internal direction of a nonzero complex triplet modulo overall phase is

$$\mathbb{CP}^2;$$

2. this is exactly the internal geometry previously identified as the correct TECT candidate for the emergent nonabelian sector;
3. the complex triplet is the minimal field that naturally supports both shell condensation and a projective internal manifold.

C. Minimal microscopic assumptions

We impose the following assumptions.

- A1. Locality.* The microscopic action is local in space and time.
- A2. Translation invariance.* No explicit x -dependence appears.
- A3. Pre-condensation isotropy.* Before shell selection, the microscopic action is $SO(3)$ -invariant in space.
- A4. Internal unitary symmetry.* At the microscopic level, the condensate field transforms as

$$\Psi \mapsto U\Psi, \quad U \in U(3),$$

or at least under the subgroup sufficient to preserve $\Psi^\dagger \Psi$.

- A5. Time-reversal/parity-even lowest-order dynamics.* The fundamental dynamics is second order in time and even under $t \mapsto -t$, $x \mapsto -x$.
- A6. Lowest-order derivative truncation.* We keep terms up to:

- quadratic order in time derivatives,
- fourth order in spatial derivatives,
- quartic order in the field amplitude.

- A7. Stable shell-forming spectrum.* The spatial derivative terms must permit a minimum at a nonzero shell q_* .

These are the weakest assumptions under which one can hope to reconstruct a usable microscopic TECT equation.

D. Reconstruction theorem for the minimal action

We now determine the most general action compatible with A1–A6.

[Minimal microscopic TECT action under the truncation assumptions] Let $\Psi : \mathbb{R}^{3,1} \rightarrow \mathbb{C}^3$ satisfy assumptions A1–A6. Then, up to total derivatives, field rescalings, and an overall normalization, the most general local $SO(3)$ -invariant, $U(3)$ -invariant action with at most two time derivatives, four spatial derivatives, and quartic field self-interaction is

$$S[\Psi] = \int dt d^3x \left[\frac{1}{2} \partial_t \Psi^\dagger \partial_t \Psi - \frac{\mu^2}{2} \Psi^\dagger \Psi + \frac{Z}{2} \partial_i \Psi^\dagger \partial_i \Psi - \frac{Y}{2} (\nabla^2 \Psi)^\dagger (\nabla^2 \Psi) - \frac{g}{4} (\Psi^\dagger \Psi)^2 \right]. \quad (11)$$

By locality and translation invariance, the Lagrangian density may depend only on Ψ , $\Psi^\dagger$, and their derivatives. By $U(3)$ -invariance, scalar terms must be built from contractions of the form

$$\Psi^\dagger \Psi, \quad \partial_\mu \Psi^\dagger \partial_\nu \Psi, \quad (\nabla^2 \Psi)^\dagger (\nabla^2 \Psi),$$

and products thereof.

By $SO(3)$ -invariance and parity evenness, the only scalar terms involving up to four spatial derivatives are, up to integrations by parts:

$$\Psi^\dagger \Psi, \quad \partial_i \Psi^\dagger \partial_i \Psi, \quad (\nabla^2 \Psi)^\dagger (\nabla^2 \Psi).$$

Mixed terms such as

$$\Psi^\dagger \nabla^2 \Psi$$

are equivalent, up to total derivatives, to $-\partial_i \Psi^\dagger \partial_i \Psi$, hence are not independent.

By time-reversal evenness and the assumption of second-order-in-time dynamics, the only independent quadratic time term is

$$\partial_t \Psi^\dagger \partial_t \Psi.$$

Terms linear in $\partial_t \Psi$ are odd under time reversal and are excluded by A5.

By quartic truncation and $U(3)$ -invariance, the only independent local quartic potential term is

$$(\Psi^\dagger \Psi)^2.$$

Therefore the action must be of the form (11).

E. Microscopic Euler–Lagrange equation

Varying (11) with respect to $\Psi^\dagger$, we obtain

$$\partial_t^2 \Psi = -\mu^2 \Psi + Z \nabla^2 \Psi - Y \nabla^4 \Psi - g(\Psi^\dagger \Psi) \Psi. \quad (12)$$

This is the minimal microscopic TECT equation in the truncation class defined above.

$$\boxed{\partial_t^2 \Psi = -\mu^2 \Psi + Z \nabla^2 \Psi - Y \nabla^4 \Psi - g(\Psi^\dagger \Psi) \Psi.} \quad (13)$$

F. Linear dispersion and shell instability

Linearize around the trivial phase $\Psi = 0$ by writing

$$\Psi(x, t) = \Psi_k e^{i(k \cdot x - \omega t)}.$$

Then (12) gives

$$\omega^2(k) = \mu^2 + Z|k|^2 + Y|k|^4. \quad (14)$$

We seek a nonzero shell minimum. Differentiating with respect to $k = |k|$,

$$\frac{d\omega^2}{dk} = 2Zk + 4Yk^3 = 2k(Z + 2Yk^2).$$

Hence the nonzero stationary shell occurs at

$$q_*^2 = -\frac{Z}{2Y}, \quad (15)$$

provided

$$Z < 0, \quad Y > 0. \quad (16)$$

Thus the finite-shell instability required by TECT is automatic in the microscopic equation (13) once (16) holds.

G. Quartic stabilization

The quartic term in the action is

$$\frac{g}{4}(\Psi^\dagger \Psi)^2.$$

Stability of the condensate amplitude requires

$$g > 0. \quad (17)$$

Thus the complete minimal shell-forming parameter regime is

$$\boxed{Z < 0, \quad Y > 0, \quad g > 0.} \quad (18)$$

H. Amplitude-projective decomposition

To expose the internal geometry, write

$$\Psi(x, t) = \rho(x, t) z(x, t), \quad \rho(x, t) \geq 0, \quad z^\dagger z = 1. \quad (19)$$

Then z is defined only up to local phase,

$$z \sim e^{i\chi(x, t)} z,$$

so the internal target space is

$$z \in \mathbb{CP}^2. \quad (20)$$

This is the first-principles microscopic origin of the TECT $\mathbb{CP}^2$ internal sector.

I. Decomposition of the quadratic kinetic terms

Using $\Psi = \rho z$, we compute

$$\partial_t \Psi = (\partial_t \rho) z + \rho \partial_t z, \quad \partial_i \Psi = (\partial_i \rho) z + \rho \partial_i z. \quad (21)$$

Since $z^\dagger z = 1$, one has

$$z^\dagger \partial_\mu z + (\partial_\mu z^\dagger) z = 0,$$

so the cross terms are purely imaginary and drop out of the norm squares. Therefore

$$\partial_t \Psi^\dagger \partial_t \Psi = (\partial_t \rho)^2 + \rho^2 \partial_t z^\dagger \partial_t z, \quad (22)$$

$$\partial_i \Psi^\dagger \partial_i \Psi = (\partial_i \rho)^2 + \rho^2 \partial_i z^\dagger \partial_i z. \quad (23)$$

Thus the lowest-order part of the action becomes

$$\begin{aligned} S[\rho, z] = \int dt d^3x & \left[\frac{1}{2}(\partial_t \rho)^2 + \frac{\rho^2}{2} \partial_t z^\dagger \partial_t z - \frac{\mu^2}{2} \rho^2 + \frac{Z}{2}(\partial_i \rho)^2 + \frac{Z\rho^2}{2} \partial_i z^\dagger \partial_i z - \frac{g}{4} \rho^4 \right] \\ & - \frac{Y}{2} \int dt d^3x (\nabla^2(\rho z))^\dagger (\nabla^2(\rho z)). \end{aligned} \quad (24)$$

This shows rigorously that the internal orientation field is projective and that the $\mathbb{CP}^2$ sector is not inserted by hand.

J. Frozen-amplitude regime and emergent $\mathbb{CP}^2$ sector

In the condensed phase, if the amplitude mode is heavy compared to the internal orientation mode, one may freeze

$$\rho(x, t) \approx \rho_0 > 0.$$

Then the internal dynamics reduces, at lowest derivative order, to a projective sigma-model type action

$$S_{\text{int}} \sim \frac{\rho_0^2}{2} \int dt d^3x \left(\partial_t z^\dagger \partial_t z + Z \partial_i z^\dagger \partial_i z \right) + \dots, \quad z^\dagger z = 1, \quad z \sim e^{iX} z. \quad (25)$$

Hence the microscopic equation (13) automatically contains a $\mathbb{CP}^2$ internal geometry after condensation.

K. Why this is the correct microscopic candidate

The field equation (13) simultaneously gives:

1. a finite-shell instability at

$$|k| = q_*;$$

2. a stable quartically bounded condensate phase;
3. a complex-triplet internal sector whose projective orientation is $\mathbb{CP}^2$;
4. a second-order time evolution compatible with a relativistic-type elastic limit after condensation.

Thus it is the unique lowest-order local microscopic candidate compatible with all currently established TECT structural requirements.

L. Honest uniqueness statement

The uniqueness proved here is not absolute UV uniqueness. It is uniqueness within the truncation class defined by A1–A7.

So the honest conclusion is:

Within the minimal local, isotropic, $U(3)$ -invariant, fourth-order spatial truncation,

(26)

the microscopic TECT equation is essentially the complex-triplet Brazovskii wave equation (13).

(27)

This is the correct next-level replacement for the previous circular shell-level discussion.

M. What is now closed and what remains open

What is now rigorously established:

1. the minimal microscopic TECT field must be a complex triplet if the internal manifold is to be $\mathbb{CP}^2$;
2. under the stated locality/symmetry/derivative assumptions, the lowest-order action is uniquely of the form (11);
3. the corresponding microscopic equation is (13);
4. the shell instability condition is

$$Z < 0, \quad Y > 0;$$

5. quartic stabilization requires

$$g > 0;$$

6. the internal projective geometry $\mathbb{CP}^2$ emerges from amplitude-direction decomposition.

What remains open:

1. whether still-higher-order operators must be included in the final UV completion;
2. whether the condensed background of (13) selects BCC dynamically for the triplet model exactly as in the scalar shell-reduced analysis;
3. whether the full shell-enriched witness data computed from (13) are nonzero;
4. whether the actual remote-sector gap and mass-protection symmetry hold in the full triplet realization.

IV. FINAL SUMMARY

Under the minimal assumptions of locality, translation invariance, pre-condensation isotropy, $U(3)$ internal invariance, second-order time dynamics, fourth-order spatial derivative truncation, and quartic stabilization, the microscopic TECT field must be taken as a complex triplet

$$\Psi(x, t) \in \mathbb{C}^3.$$

Within this truncation class, the most general microscopic action is

$$S[\Psi] = \int dt d^3x \left[\frac{1}{2} \partial_t \Psi^\dagger \partial_t \Psi - \frac{\mu^2}{2} \Psi^\dagger \Psi + \frac{Z}{2} \partial_i \Psi^\dagger \partial_i \Psi - \frac{Y}{2} (\nabla^2 \Psi)^\dagger (\nabla^2 \Psi) - \frac{g}{4} (\Psi^\dagger \Psi)^2 \right].$$

Its Euler–Lagrange equation is

$$\partial_t^2 \Psi = -\mu^2 \Psi + Z \nabla^2 \Psi - Y \nabla^4 \Psi - g (\Psi^\dagger \Psi) \Psi.$$

The linear dispersion is

$$\omega^2(k) = \mu^2 + Z|k|^2 + Y|k|^4,$$

so a finite-shell instability occurs at

$$q_*^2 = -\frac{Z}{2Y}$$

when

$$Z < 0, \quad Y > 0.$$

Quartic stabilization requires

$$g > 0.$$

Writing

$$\Psi = \rho z, \quad z^\dagger z = 1, \quad z \sim e^{i\chi} z,$$

shows that the internal orientation sector is projective:

$$z \in \mathbb{CP}^2.$$

Therefore, within the minimal local isotropic $U(3)$ -invariant truncation, the microscopic TECT equation is essentially the complex-triplet Brazovskii wave equation. This is the correct minimal microscopic candidate underlying the shell-reduced TECT fermion program.

V. STATIC FREE ENERGY OF THE COMPLEX-TRIPLET BRAZOVSKII MODEL AND THE BCC SHELL ANSATZ

A. Goal

We now return to the actual microscopic triplet model and ask the structural-selection question directly:

Does the scalar BCC-selection theorem remain valid for the complex-triplet Brazovskii model?

The answer is:

1. *Yes*, exactly, in the polarization-locked sector

$$\Psi(x) = z \phi(x), \quad z^\dagger z = 1, \quad z = \text{const};$$

2. *not yet globally proven* in the unrestricted triplet problem, where different shell modes may carry different internal polarizations.

So the correct theorem is a *lifting theorem* from the scalar shell theory to the locked triplet sector, together with an honest statement of what remains open beyond that sector.

B. Static free energy of the microscopic triplet model

Start from the minimal microscopic triplet Brazovskii equation

$$\partial_t^2 \Psi = -\mu^2 \Psi + Z \nabla^2 \Psi - Y \nabla^4 \Psi - g(\Psi^\dagger \Psi) \Psi, \quad \Psi : \mathbb{R}^{3,1} \rightarrow \mathbb{C}^3.$$

Its corresponding static free-energy functional is

$$\mathcal{F}[\Psi] = \int d^3x \left[\frac{r}{2} \Psi^\dagger \Psi + \frac{Z}{2} \partial_i \Psi^\dagger \partial_i \Psi + \frac{Y}{2} (\nabla^2 \Psi)^\dagger (\nabla^2 \Psi) + \frac{g}{4} (\Psi^\dagger \Psi)^2 \right], \quad (28)$$

with

$$Z < 0, \quad Y > 0, \quad g > 0.$$

The quadratic kernel is

$$\omega(k) = r + Z|k|^2 + Y|k|^4, \quad (29)$$

whose shell minimum occurs at

$$q_*^2 = -\frac{Z}{2Y}. \quad (30)$$

C. Polarization-locked sector

Let

$$\Psi(x) = z \phi(x), \quad z \in \mathbb{C}^3, \quad z^\dagger z = 1, \quad z = \text{const}. \quad (31)$$

Then

$$\Psi^\dagger \Psi = (z^\dagger z) |\phi|^2 = |\phi|^2, \quad (32)$$

$$\partial_i \Psi^\dagger \partial_i \Psi = (z^\dagger z) \partial_i \phi^* \partial_i \phi = |\nabla \phi|^2, \quad (33)$$

and

$$(\nabla^2 \Psi)^\dagger (\nabla^2 \Psi) = (z^\dagger z) |\nabla^2 \phi|^2 = |\nabla^2 \phi|^2. \quad (34)$$

Substituting (32)–(34) into (28), we obtain

$$\mathcal{F}[z\phi] = \int d^3x \left[\frac{r}{2} |\phi|^2 + \frac{Z}{2} |\nabla \phi|^2 + \frac{Y}{2} |\nabla^2 \phi|^2 + \frac{g}{4} |\phi|^4 \right]. \quad (35)$$

This is exactly the scalar Brazovskii free energy.

[Exact scalar reduction in the polarization-locked sector] For every constant unit vector $z \in \mathbb{C}^3$, the restriction of the triplet static free energy (28) to configurations of the form $\Psi = z\phi$ is exactly the scalar Brazovskii free energy (35).

Immediate from (32)–(34).

D. BCC shell ansatz in the locked sector

Let the BCC first shell be

$$S_{\text{BCC}} = \frac{q_*}{\sqrt{2}} \{(\pm 1, \pm 1, 0), (\pm 1, 0, \pm 1), (0, \pm 1, \pm 1)\}. \quad (36)$$

Choose one representative from each antipodal pair,

$$Q_a \in S_{\text{BCC}}, \quad a = 1, \dots, 6,$$

and define the locked BCC triplet ansatz

$$\Psi_{\text{BCC}}(x) = z \phi_{\text{BCC}}(x), \quad \phi_{\text{BCC}}(x) = 2A \sum_{a=1}^6 \cos(Q_a \cdot x), \quad z^\dagger z = 1. \quad (37)$$

By the exact scalar reduction theorem,

$$\mathcal{F}[\Psi_{\text{BCC}}] = \mathcal{F}_{\text{scalar}}[\phi_{\text{BCC}}]. \quad (38)$$

Therefore the shell-projected amplitude equation is identical to the scalar one:

$$\left(r - \frac{Z^2}{4Y}\right) A + \beta_{\text{BCC}} g A^3 = 0. \quad (39)$$

Hence

$$A^2 = -\frac{1}{\beta_{\text{BCC}} g} \left(r - \frac{Z^2}{4Y}\right). \quad (40)$$

So every scalar shell-selection result in the single-field Brazovskii theory lifts *exactly* to the locked triplet sector.

E. Immediate consequence for the BCC selection theorem

Assume the scalar single-shell Brazovskii theory has already established the ordering

$$F_{\text{BCC}}^{\min} < F_{\text{FCC}}^{\min} < F_{\text{SC}}^{\min}$$

within the relevant universality class. Then, by the exact reduction (35), the same ordering holds for the locked triplet sector:

$$F_{\text{BCC,lock}}^{\min} < F_{\text{FCC,lock}}^{\min} < F_{\text{SC,lock}}^{\min}. \quad (41)$$

[Lift of scalar BCC selection to the locked triplet sector] Within the polarization-locked sector $\Psi = z\phi$ with constant $z^\dagger z = 1$, the complex-triplet Brazovskii free energy is exactly identical to the scalar Brazovskii free energy. Consequently, every scalar shell-selection theorem — in particular the scalar BCC-selection theorem — holds verbatim in the locked triplet sector.

The first statement is the exact scalar reduction theorem. The second follows immediately because the variational problem on the locked triplet sector is identical to the scalar one.

F. General unrestricted triplet shell ansatz

We now separate what happens beyond the locked sector.
Consider the more general shell ansatz

$$\Psi(x) = \sum_{a=1}^6 (\eta_a u_a e^{iQ_a \cdot x} + \bar{\eta}_a \bar{u}_a e^{-iQ_a \cdot x}), \quad u_a^\dagger u_a = 1, \quad u_a \in \mathbb{C}^3. \quad (42)$$

Here each shell mode may carry its own internal polarization u_a .

a. Quadratic part. Because the quadratic kernel is internal-index blind and depends only on $|Q_a| = q_*$, the shell-projected quadratic free energy depends only on the amplitudes:

$$\mathcal{F}_2 = r_* \sum_{a=1}^6 |\eta_a|^2, \quad r_* := r - \frac{Z^2}{4Y}, \quad (43)$$

and is independent of the polarization vectors u_a .

b. Quartic part. The quartic term becomes

$$\mathcal{F}_4 = \frac{g}{4} \int d^3x (\Psi^\dagger \Psi)^2. \quad (44)$$

Substituting (42), the shell-projected quartic part takes the schematic form

$$\mathcal{F}_4^{\text{shell}} = g \sum_{a,b,c,d} \mathcal{C}_{abcd} (u_a^\dagger u_b) (u_c^\dagger u_d) \eta_a \bar{\eta}_b \eta_c \bar{\eta}_d, \quad (45)$$

where $\mathcal{C}_{abcd}$ is nonzero only when the shell resonance constraint

$$Q_a - Q_b + Q_c - Q_d = 0$$

is satisfied.

Thus, in the unrestricted triplet problem, the quartic coefficient depends not only on the shell geometry, but also on the internal overlap matrix

$$G_{ab} := u_a^\dagger u_b.$$

G. Locked sector as a special case

If all internal polarizations are equal,

$$u_a = z \quad \forall a, \quad z^\dagger z = 1,$$

then

$$u_a^\dagger u_b = 1 \quad \forall a, b,$$

and (45) collapses exactly to the scalar shell quartic form. Hence the scalar BCC coefficient β_{BCC} is recovered.

So the locked sector is the special case

$$G_{ab} \equiv 1.$$

H. Honest limitation

At this point an important honesty statement is required.

The scalar BCC-selection theorem lifts *exactly* to the locked sector. However, for the fully unrestricted triplet shell problem, the quartic form (45) depends on the internal overlap matrix G_{ab} , and therefore the scalar structure-selection theorem does *not* automatically imply the global triplet theorem.

In other words:

$$\boxed{\text{scalar BCC selection survives exactly in the polarization-locked triplet sector,}} \quad (46)$$

but

$$\boxed{\text{global BCC selection in the unrestricted triplet shell theory is still open.}} \quad (47)$$

This is not a failure of the model. It is the correct mathematical separation between what has been proved and what remains to be checked.

I. Sufficient condition for full triplet inheritance

There is an immediate sufficient condition under which the scalar theorem becomes the full triplet theorem.

If one proves that the triplet free energy is minimized, among shell configurations with a given geometric star, by internal polarization locking

$$u_1 = \dots = u_6 = z,$$

then the unrestricted triplet minimization reduces to the scalar one, and the scalar BCC theorem lifts globally.

Thus the remaining extra step is:

$$\boxed{\text{prove internal polarization locking in the shell-condensed phase.}} \quad (48)$$

Once (48) is established, BCC selection for the full triplet model follows immediately from the scalar theorem.

J. What is now closed and what remains open

What is now rigorously established:

1. the static free energy of the complex-triplet Brazovskii model is exactly reducible to the scalar Brazovskii free energy in the constant-polarization sector;
2. the locked BCC triplet ansatz satisfies the same amplitude equation as the scalar BCC ansatz;
3. therefore the scalar BCC-selection theorem remains exactly valid in the locked triplet sector.

What remains open:

1. whether the unrestricted triplet shell minimization dynamically locks all internal polarizations;
2. therefore, whether scalar BCC selection lifts from the locked sector to the full unrestricted triplet theory.

VI. FINAL SUMMARY

The static free energy of the complex-triplet Brazovskii model is

$$\mathcal{F}[\Psi] = \int d^3x \left[\frac{r}{2} \Psi^\dagger \Psi + \frac{Z}{2} \partial_i \Psi^\dagger \partial_i \Psi + \frac{Y}{2} (\nabla^2 \Psi)^\dagger (\nabla^2 \Psi) + \frac{g}{4} (\Psi^\dagger \Psi)^2 \right], \quad Z < 0, \ Y > 0, \ g > 0.$$

If one restricts to the polarization-locked sector

$$\Psi(x) = z \phi(x), \quad z^\dagger z = 1, \quad z = \text{const},$$

then

$$\mathcal{F}[z\phi] = \int d^3x \left[\frac{r}{2} |\phi|^2 + \frac{Z}{2} |\nabla \phi|^2 + \frac{Y}{2} |\nabla^2 \phi|^2 + \frac{g}{4} |\phi|^4 \right],$$

which is exactly the scalar Brazovskii free energy.

Therefore, for the locked BCC shell ansatz

$$\Psi_{\text{BCC}}(x) = z \phi_{\text{BCC}}(x), \quad \phi_{\text{BCC}}(x) = 2A \sum_{a=1}^6 \cos(Q_a \cdot x),$$

the amplitude equation is identical to the scalar one:

$$\left(r - \frac{Z^2}{4Y}\right) A + \beta_{\text{BCC}} g A^3 = 0.$$

Hence the scalar BCC-selection theorem remains exactly valid in the polarization-locked triplet sector. However, in the unrestricted triplet shell ansatz

$$\Psi(x) = \sum_{a=1}^6 (\eta_a u_a e^{iQ_a \cdot x} + \bar{\eta}_a \bar{u}_a e^{-iQ_a \cdot x}),$$

the quartic shell energy depends on the internal overlap matrix

$$G_{ab} = u_a^\dagger u_b,$$

so the scalar theorem does not yet automatically imply the full global triplet theorem.

Thus the exact current status is:

BCC selection is rigorously inherited in the polarization-locked triplet sector,

but

global BCC selection for the unrestricted triplet model remains open until internal polarization locking is proved.

VII. QUARTIC SHELL ENERGY OF THE UNRESTRICTED COMPLEX-TRIPLET BRAZOVSKII MODEL AND THE POLARIZATION-LOCKING QUESTION

A. Goal

We now study the unrestricted triplet shell ansatz and ask whether the quartic shell energy by itself favors internal polarization locking.

The answer is:

No: the quartic term alone does not generically force polarization locking. (49)

More precisely, the quadratic shell energy is polarization-blind, while the quartic energy contains positive cross terms weighted by internal overlaps. Therefore reducing the overlaps can lower the quartic coefficient. Thus the locked sector is not automatically the global minimizer of the unrestricted triplet shell problem.

B. Static free energy

The static free energy of the complex-triplet Brazovskii model is

$$\mathcal{F}[\Psi] = \int d^3x \left[\frac{r}{2} \Psi^\dagger \Psi + \frac{Z}{2} \partial_i \Psi^\dagger \partial_i \Psi + \frac{Y}{2} (\nabla^2 \Psi)^\dagger (\nabla^2 \Psi) + \frac{g}{4} (\Psi^\dagger \Psi)^2 \right], \quad (50)$$

with

$$Z < 0, \quad Y > 0, \quad g > 0.$$

The shell radius is

$$q_*^2 = -\frac{Z}{2Y}.$$

C. Unrestricted shell ansatz

Let S denote the selected shell, for example the BCC first shell. The most general shell ansatz is

$$\Psi(x) = \sum_{G \in S} \psi_G u_G e^{iG \cdot x}, \quad u_G^\dagger u_G = 1, \quad u_G \in \mathbb{C}^3. \quad (51)$$

Here:

- $\psi_G \in \mathbb{C}$ are shell amplitudes,
- $u_G \in \mathbb{C}^3$ are internal polarization vectors.

Unlike the locked sector, we do *not* assume u_G is the same for all shell modes.

D. Quadratic shell energy is polarization-blind

Substituting (51) into the quadratic part of (50), we obtain

$$\mathcal{F}_2 = \frac{1}{2} \sum_{G \in S} (r + Z|G|^2 + Y|G|^4) |\psi_G|^2 (u_G^\dagger u_G). \quad (52)$$

Since $u_G^\dagger u_G = 1$ and $|G| = q_*$ on the shell,

$$\mathcal{F}_2 = \frac{r_*}{2} \sum_{G \in S} |\psi_G|^2, \quad r_* := r + Zq_*^2 + Yq_*^4 = r - \frac{Z^2}{4Y}. \quad (53)$$

Therefore:

the shell-projected quadratic energy is completely independent of the internal polarizations u_G .

(54)

So polarization selection, if any, must come from the quartic term.

E. Quartic shell energy

We now compute the quartic contribution. First,

$$\Psi^\dagger \Psi = \sum_{G, H \in S} \bar{\psi}_G \psi_H (u_G^\dagger u_H) e^{i(H-G) \cdot x}. \quad (55)$$

Hence

$$(\Psi^\dagger \Psi)^2 = \sum_{G, H, K, L \in S} \bar{\psi}_G \psi_H \bar{\psi}_K \psi_L (u_G^\dagger u_H) (u_K^\dagger u_L) e^{i(H-G+L-K) \cdot x}. \quad (56)$$

Integrating over space, only resonant quartets survive:

$$H - G + L - K = 0.$$

Equivalently,

$$G + K = H + L.$$

Therefore the shell-projected quartic energy is

$$\mathcal{F}_4^{\text{shell}} = \frac{gV}{4} \sum_{\substack{G, H, K, L \in S \\ G+K=H+L}} \bar{\psi}_G \psi_H \bar{\psi}_K \psi_L (u_G^\dagger u_H) (u_K^\dagger u_L),$$

(57)

where V is the spatial volume.

This is the exact quartic shell functional for the unrestricted triplet shell ansatz.

F. Locked sector as the scalar reduction

If

$$u_G = z \quad \forall G \in S, \quad z^\dagger z = 1,$$

then

$$u_G^\dagger u_H = 1 \quad \forall G, H,$$

and (57) reduces exactly to the scalar shell quartic form. Thus the scalar BCC theorem is recovered exactly in the locked sector.

However, the unrestricted problem is controlled by the Gram matrix

$$\Gamma_{GH} := u_G^\dagger u_H, \quad (58)$$

which is no longer identically 1.

G. Diagonal quartic cross terms

To determine whether locking is favored, it suffices to examine a universal positive subset of the quartic terms. Choose the resonant quartets

$$H = G, \quad L = K.$$

These always satisfy $G + K = H + L$. Their contribution to (57) is

$$\mathcal{F}_{4,\text{diag}} = \frac{gV}{4} \sum_{G,K \in S} |\psi_G|^2 |\psi_K|^2 (u_G^\dagger u_G)(u_K^\dagger u_K) = \frac{gV}{4} \left(\sum_{G \in S} |\psi_G|^2 \right)^2. \quad (59)$$

This part is polarization-independent.

But there is another universal class:

$$H = K, \quad L = G,$$

which also always satisfies $G + K = H + L$. Its contribution is

$$\mathcal{F}_{4,\text{ex}} = \frac{gV}{4} \sum_{G,K \in S} \bar{\psi}_G \psi_K \bar{\psi}_K \psi_G (u_G^\dagger u_K)(u_K^\dagger u_G). \quad (60)$$

Since

$$\bar{\psi}_G \psi_K \bar{\psi}_K \psi_G = |\psi_G|^2 |\psi_K|^2, \quad (u_G^\dagger u_K)(u_K^\dagger u_G) = |u_G^\dagger u_K|^2,$$

we obtain

$$\boxed{\mathcal{F}_{4,\text{ex}} = \frac{gV}{4} \sum_{G,K \in S} |\psi_G|^2 |\psi_K|^2 |u_G^\dagger u_K|^2.} \quad (61)$$

This contribution is manifestly nonnegative and depends monotonically on the pairwise overlaps.

H. Immediate consequence

Because

$$0 \leq |u_G^\dagger u_K|^2 \leq 1,$$

the exchange part (61) is minimized by reducing internal overlaps and is maximized when all polarizations are parallel:

$$u_G = u_K \quad \implies \quad |u_G^\dagger u_K|^2 = 1.$$

Thus, at fixed shell amplitudes $|\psi_G|$,

the universal overlap-dependent quartic contribution is minimized by reducing polarization overlaps,

(62)

not by locking them.

Therefore the quartic term alone does not energetically force internal polarization locking.

I. Two-mode example

The above conclusion is already visible in the simplest nontrivial example.

Take two nonzero shell modes G_1, G_2 with amplitudes ψ_1, ψ_2 and internal polarizations u_1, u_2 . Restricting to the exchange contribution (61), one finds

$$\mathcal{F}_{4,\text{ex}}^{(2\text{-mode})} = \frac{gV}{4} \left(|\psi_1|^4 + |\psi_2|^4 + 2|\psi_1|^2 |\psi_2|^2 |u_1^\dagger u_2|^2 \right). \quad (63)$$

At fixed amplitudes, this is minimized when $|u_1^\dagger u_2|^2$ is as small as possible and maximized when $u_1 \parallel u_2$.

So already in the simplest nontrivial shell configuration, locking is not favored by the quartic term.

J. Consequence for the unrestricted minimization problem

Since the quadratic shell energy is polarization-blind and the quartic energy contains positive overlap terms such as (61), there is no general mechanism in the minimal triplet Brazovskii free energy that forces all shell polarizations to align.

Hence the following statement is false in general:

“The unrestricted triplet shell minimizer is automatically polarization-locked.”

Instead, the correct theorem-level statement is:

The scalar BCC-selection theorem lifts exactly to the locked sector, but the unrestricted triplet global minimizer is not yet determined.

(64)

K. What would be needed to recover global locking

To make the full triplet theory inherit scalar BCC selection globally, one would need an additional mechanism favoring

$$u_G = u_H \quad \forall G, H \in S.$$

Such a mechanism could come from:

1. additional microscopic alignment terms,
2. derivative couplings penalizing internal relative misalignment,
3. gauge/projector-sector effects not visible in the bare amplitude-only quartic term,
4. topological or defect constraints in the condensed phase.

But none of these follows from the minimal quartic triplet free energy (50) alone.

L. The correct theorem

We summarize the result precisely.

[Quartic term does not force polarization locking] Consider the unrestricted shell ansatz

$$\Psi(x) = \sum_{G \in S} \psi_G u_G e^{iG \cdot x}, \quad u_G^\dagger u_G = 1,$$

for the complex-triplet Brazovskii free energy (50). Then:

1. the shell-projected quadratic energy is independent of the internal polarizations u_G ;
2. the shell-projected quartic energy contains the universal positive exchange term

$$\frac{gV}{4} \sum_{G, K \in S} |\psi_G|^2 |\psi_K|^2 |u_G^\dagger u_K|^2;$$

3. therefore the quartic term alone does not favor global polarization locking, and can instead favor reducing pairwise overlaps.

Item (1) follows from (53). Item (2) follows from the resonant quartets $H = K$, $L = G$ in (57). Item (3) follows because $|u_G^\dagger u_K|^2 \in [0, 1]$, so the exchange term is minimized by reducing overlaps and maximized by locking.

M. Final honest status

What is now rigorously established:

1. the polarization-locked triplet sector is exactly equivalent to the scalar Brazovskii theory;
2. the unrestricted triplet quartic shell energy depends on the internal Gram matrix $u_G^\dagger u_H$;
3. the universal positive overlap terms show that quartic energy alone does not force polarization locking.

What remains open:

1. what additional microscopic mechanism aligns the internal polarizations, if any;
2. whether the actual TECT microscopic completion contains such an alignment mechanism;
3. whether global BCC selection survives in the full unrestricted triplet theory after those extra mechanisms are included.

VIII. FINAL SUMMARY

For the unrestricted shell ansatz

$$\Psi(x) = \sum_{G \in S} \psi_G u_G e^{iG \cdot x}, \quad u_G^\dagger u_G = 1,$$

the quadratic shell energy of the complex-triplet Brazovskii model is polarization-blind:

$$\mathcal{F}_2 = \frac{r_*}{2} \sum_{G \in S} |\psi_G|^2, \quad r_* = r - \frac{Z^2}{4Y}.$$

The quartic shell energy is

$$\mathcal{F}_4^{\text{shell}} = \frac{gV}{4} \sum_{\substack{G, H, K, L \in S \\ G+K=H+L}} \bar{\psi}_G \psi_H \bar{\psi}_K \psi_L (u_G^\dagger u_H) (u_K^\dagger u_L).$$

It contains the universal positive exchange contribution

$$\frac{gV}{4} \sum_{G,K \in S} |\psi_G|^2 |\psi_K|^2 |u_G^\dagger u_K|^2.$$

Since

$$0 \leq |u_G^\dagger u_K|^2 \leq 1,$$

this contribution is minimized by reducing polarization overlaps and maximized by full locking. Therefore the quartic term alone does not force internal polarization locking.

Hence the exact current status is:

the scalar BCC-selection theorem remains exactly valid in the polarization-locked triplet sector,

but

the unrestricted triplet global minimization problem remains open, because the quartic term alone does not select locking.

IX. A NATURAL ALIGNMENT MECHANISM FOR THE TRIPLET BRAZOVSKII MODEL

A. Goal

The quartic shell analysis showed that the bare complex-triplet Brazovskii free energy does *not* automatically force internal polarization locking. Therefore, if global BCC selection is to be inherited from the scalar theorem, the microscopic TECT model must contain an additional alignment mechanism.

The purpose of this section is to identify a natural extra term that:

1. favors internal polarization locking;
2. preserves the already-established $\mathbb{CP}^2$ internal geometry;
3. does not spoil the shell/BCC/Weyl/Dirac framework already derived;
4. gives a mathematically controlled route from the unrestricted triplet problem back to the locked sector.

B. Amplitude-projective variables

Write the triplet field as

$$\Psi(x) = \rho(x) z(x), \quad \rho(x) \geq 0, \quad z^\dagger z = 1, \quad z \sim e^{i\chi(x)} z. \quad (65)$$

Then z is a local representative of a point in

$$\mathbb{CP}^2.$$

Define the rank-one projector

$$P(x) := z(x) z^\dagger(x), \quad P^2 = P, \quad P^\dagger = P, \quad \text{Tr } P = 1. \quad (66)$$

Define also the projective covariant derivative

$$D_i z := (1 - P) \partial_i z. \quad (67)$$

This is invariant under the local phase redundancy $z \sim e^{i\chi} z$, and one has the standard identity

$$\text{Tr}(\partial_i P \partial_i P) = 2 (D_i z)^\dagger (D_i z). \quad (68)$$

C. Natural alignment term

The most natural local, rotationally invariant, phase-redundancy-invariant, lowest-order alignment term is

$$\boxed{\mathcal{F}_{\text{align}}[\rho, z] = \frac{\chi}{2} \int d^3x \rho^2 (D_i z)^\dagger (D_i z), \quad \chi > 0.} \quad (69)$$

Equivalently, using (68),

$$\boxed{\mathcal{F}_{\text{align}}[\rho, P] = \frac{\chi}{4} \int d^3x \rho^2 \text{Tr}(\partial_i P \partial_i P).} \quad (70)$$

This term is positive semidefinite and penalizes spatial variation of the internal projective orientation.

D. Why this term is natural

The alignment term (69) is natural because it satisfies all structural requirements:

1. it is local;
2. it is translationally invariant;
3. it is rotationally invariant;
4. it preserves the projective $U(1)$ redundancy of z ;
5. it depends only on the $\mathbb{CP}^2$ orientation field and the condensate amplitude;
6. it vanishes when the internal orientation is spatially constant.

Thus it is the minimal positive-definite stiffness term for the projective internal orientation.

E. Nonnegativity and zero set

We first record the most important property.

[Alignment energy is nonnegative] For every admissible configuration (ρ, z) ,

$$\mathcal{F}_{\text{align}}[\rho, z] \geq 0. \quad (71)$$

Moreover,

$$\mathcal{F}_{\text{align}}[\rho, z] = 0 \iff D_i z = 0 \quad \text{on the support of } \rho. \quad (72)$$

Equivalently,

$$\mathcal{F}_{\text{align}}[\rho, P] = 0 \iff \partial_i P = 0 \quad \text{on the support of } \rho. \quad (73)$$

Nonnegativity is immediate from (69), since $\rho^2 \geq 0$ and $(D_i z)^\dagger (D_i z) \geq 0$. Vanishing holds if and only if $D_i z = 0$ wherever $\rho \neq 0$, equivalently $\partial_i P = 0$ by (68).

Thus the alignment term is minimized exactly by projectively constant internal orientation.

F. Projectively constant orientation and shell locking

We now connect $\partial_i P = 0$ to shell polarization locking.
Consider the unrestricted shell ansatz

$$\Psi(x) = \sum_{G \in S} \psi_G u_G e^{iG \cdot x}, \quad u_G^\dagger u_G = 1, \quad u_G \in \mathbb{C}^3. \quad (74)$$

[Projective constancy implies collinearity of shell polarizations] Suppose $\Psi(x) \neq 0$ on a nonempty open set and the associated projector $P(x)$ is constant there:

$$P(x) \equiv P_0.$$

Then all shell polarization vectors u_G lie in the same complex line. Equivalently, there exists a fixed unit vector $z \in \mathbb{C}^3$ and phases $e^{i\varphi_G}$ such that

$$u_G = e^{i\varphi_G} z \quad \forall G \in S. \quad (75)$$

If $P(x) \equiv P_0$, then for all x with $\Psi(x) \neq 0$, the vector $\Psi(x)$ lies in the one-dimensional image of P_0 , i.e. in a fixed complex line $L \subset \mathbb{C}^3$. Since $\Psi(x)$ is a finite Fourier sum

$$\Psi(x) = \sum_G \psi_G u_G e^{iG \cdot x},$$

all of its Fourier coefficients must also lie in the same line L , otherwise the image of $\Psi(x)$ would span a larger subspace as x varies. Hence each $u_G \in L$, so there exists a unit vector z spanning L and phases $e^{i\varphi_G}$ such that $u_G = e^{i\varphi_G} z$.

Therefore, projective constancy is exactly the geometric content of shell polarization locking (up to irrelevant shell-dependent phases, which can be absorbed into ψ_G).

G. Locked sector remains untouched

Suppose the shell configuration is fully locked:

$$u_G = e^{i\varphi_G} z \quad \forall G.$$

Then

$$\Psi(x) = z \phi(x), \quad P(x) = z z^\dagger = \text{const},$$

so

$$\mathcal{F}_{\text{align}}[\rho, z] = 0.$$

Hence the locked sector is unaffected by the new term.

[The alignment term does not spoil the locked scalar reduction] On the polarization-locked sector $\Psi = z\phi$ with constant $z^\dagger z = 1$, one has

$$\mathcal{F}_{\text{align}} = 0.$$

Therefore the refined free energy

$$\mathcal{F}_{\text{ref}} := \mathcal{F}_{\text{triplet}} + \mathcal{F}_{\text{align}}$$

restricts exactly to the same scalar Brazovskii free energy as before.

If z is constant, then $P = z z^\dagger$ is constant, so $\partial_i P = 0$, hence $\mathcal{F}_{\text{align}} = 0$.

This is crucial: the alignment term improves the unrestricted triplet problem without changing the already-established scalar reduction on the locked sector.

H. Why the term does not spoil the shell/Weyl/Dirac framework

We now make precise why $\mathcal{F}_{\text{align}}$ is compatible with the previously developed TECT program.

a. (i) Shell instability. The shell-forming dispersion is determined by the amplitude sector of the quadratic operator

$$r + Z|k|^2 + Y|k|^4.$$

On a locked or projectively constant background, $\mathcal{F}_{\text{align}} = 0$, so the shell radius

$$q_*^2 = -\frac{Z}{2Y}$$

is unchanged at leading order.

b. (ii) BCC selection in the locked sector. Because $\mathcal{F}_{\text{align}} = 0$ on the locked sector, the scalar BCC-selection theorem remains exactly valid there.

c. (iii) Internal $\mathbb{CP}^2$ geometry. The new term is written directly in projective variables P or $D_i z$, so it preserves rather than destroys the $\mathbb{CP}^2$ structure.

d. (iv) Weyl/Dirac shell reduction. The alignment term is derivative in the internal projector. Therefore it does not introduce a constant intervalley mass term at the level of the local doubled block. Instead, it supplies positive internal-orientation stiffness and is compatible with the shell-enriched projector framework.

Thus the term (69) is structurally safe.

I. Refined microscopic TECT free energy

We therefore propose the refined static microscopic TECT free energy

$$\boxed{\mathcal{F}_{\text{ref}}[\Psi] = \mathcal{F}_{\text{triplet}}[\Psi] + \mathcal{F}_{\text{align}}[\rho, z]}, \quad (76)$$

where

$$\mathcal{F}_{\text{triplet}}[\Psi] = \int d^3x \left[\frac{r}{2} \Psi^\dagger \Psi + \frac{Z}{2} \partial_i \Psi^\dagger \partial_i \Psi + \frac{Y}{2} (\nabla^2 \Psi)^\dagger (\nabla^2 \Psi) + \frac{g}{4} (\Psi^\dagger \Psi)^2 \right], \quad (77)$$

and

$$\mathcal{F}_{\text{align}}[\rho, z] = \frac{\chi}{2} \int d^3x \rho^2 (D_i z)^\dagger (D_i z). \quad (78)$$

This is the most natural next-level refinement of the minimal triplet Brazovskii model.

J. Large- χ locking principle

We now formulate the precise variational consequence of adding $\mathcal{F}_{\text{align}}$.

Let $\mathcal{M}_R$ be any finite shell-truncated configuration manifold with fixed shell norm

$$\sum_{G \in S} |\psi_G|^2 = R^2,$$

and let $\mathcal{L}_R \subset \mathcal{M}_R$ denote the locked submanifold consisting of shell configurations projectively equivalent to

$$u_G = e^{i\varphi_G} z.$$

Define the truncated refined energy

$$E_\chi = E_0 + \chi A,$$

where E_0 is the bare triplet shell energy and $A \geq 0$ is the alignment functional restricted to $\mathcal{M}_R$.

Then $A = 0$ exactly on $\mathcal{L}_R$.

[Large- χ locking principle] Let $\mathcal{M}_R$ be compact. For every $\varepsilon > 0$, there exists $\chi_\varepsilon > 0$ such that every minimizer of E_χ on $\mathcal{M}_R$ lies within the ε -neighborhood of the locked manifold $\mathcal{L}_R$ whenever $\chi > \chi_\varepsilon$.

Since $\mathcal{M}_R$ is compact and A is continuous with zero set exactly $\mathcal{L}_R$, the complement

$$\mathcal{M}_R \setminus U_\varepsilon(\mathcal{L}_R)$$

is compact and A has a strictly positive minimum there:

$$m_\varepsilon := \min_{\mathcal{M}_R \setminus U_\varepsilon(\mathcal{L}_R)} A > 0.$$

Also, E_0 is continuous on the compact set $\mathcal{M}_R$, hence bounded:

$$|E_0| \leq C_R.$$

If $X \notin U_\varepsilon(\mathcal{L}_R)$, then

$$E_\chi(X) \geq -C_R + \chi m_\varepsilon.$$

On the locked manifold, $A = 0$, so for some locked configuration $Y \in \mathcal{L}_R$,

$$E_\chi(Y) = E_0(Y) \leq C_R.$$

Thus if

$$\chi > \frac{2C_R}{m_\varepsilon},$$

then

$$E_\chi(X) > E_\chi(Y)$$

for every $X \notin U_\varepsilon(\mathcal{L}_R)$. Hence every minimizer lies in $U_\varepsilon(\mathcal{L}_R)$.

This theorem shows that the refined model drives the unrestricted shell minimization arbitrarily close to the locked sector once χ is sufficiently large.

K. Consequence for BCC selection

Because the locked sector is exactly scalar Brazovskii, and because the large- χ minimizers are driven arbitrarily close to the locked manifold, the refined triplet model asymptotically inherits scalar BCC selection.

More precisely:

[Asymptotic lift of scalar BCC selection] Assume the scalar shell-selection theorem gives a strict BCC advantage over competing shell geometries on the locked manifold. Then, for sufficiently large alignment stiffness χ , the global minimizers of the refined triplet shell problem lie arbitrarily close to the locked manifold and therefore inherit the scalar BCC selection asymptotically.

Combine the large- χ locking principle with the exact reduction of the locked sector to the scalar Brazovskii theory.

This is the correct rigorous statement at the present stage. It is stronger than the bare triplet model result, yet still honest about what has and has not been fully proved.

L. What is now closed and what remains open

What is now rigorously established:

1. the bare quartic triplet term does not force locking;
2. the natural positive-definite projective stiffness term $\mathcal{F}_{\text{align}}$ vanishes exactly on the locked sector;
3. it preserves the scalar reduction and therefore does not spoil the existing shell/BCC/Weyl/Dirac framework;
4. in finite shell truncations, sufficiently large χ drives minimizers arbitrarily close to the locked manifold.

What remains open:

1. whether the actual microscopic TECT realization generates this projector-stiffness term from more primitive dynamics;
2. whether locking can be proved exactly (not only asymptotically) at finite χ ;
3. the full witness/gap/mass verification in the refined microscopic model.

X. FINAL SUMMARY

The bare complex-triplet Brazovskii model does not automatically favor internal polarization locking. To obtain locking without spoiling the already-established shell/BCC/Weyl/Dirac framework, the natural refinement is to add the positive projective-orientation stiffness term

$$\mathcal{F}_{\text{align}} = \frac{\chi}{2} \int d^3x \rho^2 (D_i z)^\dagger (D_i z) = \frac{\chi}{4} \int d^3x \rho^2 \text{Tr}(\partial_i P \partial_i P), \quad \chi > 0,$$

where

$$\Psi = \rho z, \quad z^\dagger z = 1, \quad P = z z^\dagger.$$

This term is nonnegative and vanishes exactly when the projective orientation is spatially constant:

$$\mathcal{F}_{\text{align}} = 0 \iff \partial_i P = 0.$$

For shell configurations, this implies that all shell polarization vectors are collinear up to phase, i.e. the configuration is locked.

Because the alignment term vanishes on the locked sector, it does not alter the exact scalar reduction there. Hence the scalar BCC-selection theorem remains intact on the locked manifold.

Moreover, in any finite shell truncation, the refined energy

$$E_\chi = E_0 + \chi A$$

has the property that, for sufficiently large χ , all minimizers lie arbitrarily close to the locked manifold. Thus the refined triplet model asymptotically inherits scalar BCC selection.

Therefore the most natural next-level microscopic TECT candidate is not the bare complex-triplet Brazovskii model alone, but the refined model

$$\mathcal{F}_{\text{ref}} = \mathcal{F}_{\text{triplet}} + \mathcal{F}_{\text{align}},$$

namely the complex-triplet Brazovskii model supplemented by projective internal-orientation stiffness.

XI. DOES THE PROJECTOR STIFFNESS TERM NEED TO BE COMPUTED FOR N_{ij}^a , THE WITNESS TRIPLE, AND THE WEYL/DIRAC CRITERIA?

A. Answer

Yes, ultimately it does need to be computed if one wants to claim an actual microscopic TECT realization. However, one must first distinguish two logically different questions:

1. Does the projector-stiffness term contribute *directly at tree level* to the enriched shell tensor

$$N_{ij}^a$$

on the locked background?

2. Does it contribute *indirectly* after the heavy internal projector fluctuations are integrated out?

The correct answer is:

On a locked constant-projector background, the projector stiffness does not generically generate a new tree-level N_{ij}^a .
(79)

But:

It does modify the effective low-energy theory indirectly by making projector fluctuations heavy and generating suppressed corrections.
(80)

Thus the computation is not optional in the final microscopic program, but its role is primarily indirect rather than direct.

B. Refined microscopic free energy

Let

$$\Psi = \rho z, \quad z^\dagger z = 1, \quad P = z z^\dagger, \quad (81)$$

and consider the refined static free energy

$$\mathcal{F}_{\text{ref}} = \mathcal{F}_{\text{triplet}} + \mathcal{F}_{\text{align}}, \quad (82)$$

with

$$\mathcal{F}_{\text{triplet}} = \int d^3x \left[\frac{r}{2} \Psi^\dagger \Psi + \frac{Z}{2} \partial_i \Psi^\dagger \partial_i \Psi + \frac{Y}{2} (\nabla^2 \Psi)^\dagger (\nabla^2 \Psi) + \frac{g}{4} (\Psi^\dagger \Psi)^2 \right], \quad (83)$$

and

$$\mathcal{F}_{\text{align}} = \frac{\chi}{4} \int d^3x \rho^2 (\partial_i P \partial_i P) = \frac{\chi}{2} \int d^3x \rho^2 (D_i z)^\dagger (D_i z). \quad (84)$$

We now ask how $\mathcal{F}_{\text{align}}$ affects the shell-enriched tensor N_{ij}^a , the witness triple $(W_\parallel, W_I, W_J)$, and the Weyl/Dirac criteria.

C. Locked background and fluctuation expansion

Let the background be polarization-locked:

$$\Psi_0(x) = \rho_0(x) z_0, \quad z_0^\dagger z_0 = 1, \quad z_0 = \text{const}, \quad P_0 = z_0 z_0^\dagger = \text{const}. \quad (85)$$

Then

$$\partial_i P_0 = 0, \quad D_i z_0 = 0.$$

Hence

$$\mathcal{F}_{\text{align}}[\Psi_0] = 0. \quad (86)$$

Now expand the projector/orientation sector:

$$z = z_0 + \pi + O(\pi^2), \quad z_0^\dagger \pi = 0, \quad (87)$$

or equivalently

$$P = P_0 + \delta P + O((\delta P)^2), \quad \delta P = z_0 \pi^\dagger + \pi z_0^\dagger. \quad (88)$$

Then

$$\partial_i P = \partial_i \delta P + O(\pi \partial \pi, \pi^2),$$

so to quadratic order

$$\mathcal{F}_{\text{align}}^{(2)} = \frac{\chi}{4} \int d^3x \rho_0(x)^2 (\partial_i \delta P \partial_i \delta P). \quad (89)$$

Thus the stiffness term begins at quadratic order in projector fluctuations around the locked background.

D. Immediate consequence: no generic tree-level contribution to N_{ij}^a

Recall that the enriched shell tensor N_{ij}^a is the coefficient of a term of schematic form

$$\delta\mathcal{L}_{\text{enr}} = N_{ij}^a (\partial_i \phi_0) (-i\partial_j) \otimes s_a, \quad (90)$$

namely a term *linear* in the shell background gradient and *linear* in the test derivative.

Now observe that $\mathcal{F}_{\text{align}}$ contains only

$$\rho^2 (\partial P \partial P),$$

which, expanded about a locked constant projector background, has:

- no term linear in δP , because $\partial_i P_0 = 0$;
- no term linear in the shell gradient $\partial_i \phi_0$ times a single derivative acting on the test mode;
- only quadratic-or-higher dependence on projector fluctuations.

Therefore:

for a locked constant-projector background, $\mathcal{F}_{\text{align}}$ does not generically create a new tree-level contribution to N_{ij}^a .
--

(91)

This is the first key fact.

E. What the stiffness term actually does at leading order

Although it does not directly create a tree-level enriched tensor on the locked background, the stiffness term does something very important: it gives the projector fluctuation π a large positive quadratic operator.

In the constant-amplitude approximation,

$$\mathcal{F}_{\text{align}}^{(2)} \approx \frac{\chi \bar{\rho}^2}{2} \int d^3x \partial_i \pi^\dagger \partial_i \pi, \quad (92)$$

so the tangent projector mode is governed by

$$\mathcal{L}_\pi \sim \chi \bar{\rho}^2 (-\nabla^2) \quad (\text{plus subleading terms}). \quad (93)$$

At shell momentum $|k| \sim q_*$, this gives a characteristic inverse propagator scale

$$\mathcal{G}_\pi^{-1}(q_*) \sim \chi \bar{\rho}^2 q_*^2. \quad (94)$$

Hence projector fluctuations become heavy when $\chi \bar{\rho}^2 q_*^2$ is large.

F. Indirect effect on N_{ij}^a : integrating out the heavy projector mode

Suppose the microscopic theory contains a coupling between shell/background fluctuations and the projector tangent mode, schematically of the form

$$\mathcal{L}_{\text{mix}} = J_a^i[\phi, \psi] \pi_i^a \quad (95)$$

or, more generally,

$$\mathcal{L}_{\text{mix}} = \mathcal{J}[\phi, \psi] \cdot \pi, \quad (96)$$

where ϕ denotes the condensate background sector and ψ denotes the shell-reduced fermionic/internal test sector.

Then eliminating π at quadratic order gives the effective contribution

$$\delta\mathcal{L}_{\text{eff}} = -\frac{1}{2} \mathcal{J} \mathcal{G}_\pi \mathcal{J} \quad (97)$$

with

$$\mathcal{G}_\pi \sim (\chi \bar{\rho}^2 (-\nabla^2) + \dots)^{-1}.$$

At shell scale $|k| \sim q_*$, this gives schematically

$$\delta \mathcal{L}_{\text{eff}} = O\left(\frac{1}{\chi \bar{\rho}^2 q_*^2}\right) \times (\text{quadratic form in the source } \mathcal{J}). \quad (98)$$

Therefore the stiffness term can induce a correction

$$\delta N_{ij}^a = O\left(\frac{1}{\chi \bar{\rho}^2 q_*^2}\right) \quad (99)$$

provided the mixing source $\mathcal{J}$ contains the correct shell-gradient/test-derivative structure.

This is the second key fact:

$\mathcal{F}_{\text{align}}$ affects N_{ij}^a indirectly, through heavy-projector-mode elimination.

(100)

G. Effect on the witness triple

The witness functionals are linear contractions of N_{ij}^a :

$$W_{\parallel}[N] = \hat{n}_i \hat{n}_j N_{ij}^3, \quad (101)$$

$$W_I[N] = \frac{1}{2} (e_{1j} \hat{n}_i N_{ij}^1 + e_{2j} \hat{n}_i N_{ij}^2), \quad (102)$$

$$W_J[N] = \frac{1}{2} (e_{1j} \hat{n}_i N_{ij}^2 - e_{2j} \hat{n}_i N_{ij}^1). \quad (103)$$

Hence if

$$N_{ij}^a \mapsto N_{ij}^a + \delta N_{ij}^a,$$

then automatically

$$W_a \mapsto W_a + \delta W_a, \quad \delta W_a = O\left(\frac{1}{\chi \bar{\rho}^2 q_*^2}\right), \quad a \in \{\parallel, I, J\}, \quad (104)$$

again assuming the heavy-mode elimination generates the corresponding tensor structure.

Thus the stiffness term does not redefine the witness concept. It renormalizes the numerical witness values through induced effective couplings.

H. Effect on the Weyl criterion

The local Weyl determinant is

$$\det M = (\kappa_* |G_*|)^3 W_{\parallel} (W_I^2 + W_J^2). \quad (105)$$

Therefore the induced shift in the determinant is

$$\delta(\det M) = (\kappa_* |G_*|)^3 [\delta W_{\parallel} (W_I^2 + W_J^2) + 2W_{\parallel} (W_I \delta W_I + W_J \delta W_J)] + O((\delta W)^2). \quad (106)$$

Hence:

- if the bare locked-sector witnesses are already nonzero, the stiffness-induced correction only renormalizes the Weyl data and does not destroy Weyl nondegeneracy for sufficiently large χ ;
- if some witness vanishes at leading order, the stiffness-induced effective correction could in principle generate a nonzero witness at subleading order, but this requires an explicit model-dependent calculation of the mixing source $\mathcal{J}$.

So the stiffness term can either stabilize an already-nonzero Weyl structure or, in principle, induce a subleading one, but only through explicit heavy-mode elimination.

I. Effect on opposite-valley matching and Dirac pairing

Because the stiffness term is written in terms of the projector P and its derivatives, it is even under $G_* \leftrightarrow -G_*$ at the locked-background level. Therefore it does not by itself generate an explicit asymmetry between the two opposite shell points.

Hence, at tree level on a matched locked background:

$$W_{\parallel}^- = W_{\parallel}^+, \quad W_I^- = W_I^+, \quad W_J^- = W_J^+. \quad (107)$$

Any mismatch can arise only through induced subleading corrections from integrating out projector fluctuations. If the underlying projector sector itself respects the opposite-valley symmetry, then these induced corrections are matched as well, so Dirac pairing survives.

Thus the stiffness term tends to *preserve* opposite-valley matching rather than spoil it.

J. Effect on the remote gap

Here the effect is much more direct.

The projector stiffness gives positive quadratic energy to projector tangent modes:

$$\mathcal{G}_{\pi}^{-1}(q_*) \sim \chi \bar{\rho}^2 q_*^2.$$

These modes are orthogonal to the locked shell-relevant Dirac block, so they belong naturally to the remote sector. Therefore the remote spectrum is shifted upward by an amount of order

$$\delta\Delta_{\text{remote}} \sim +\chi \bar{\rho}^2 q_*^2. \quad (108)$$

So even if its effect on N_{ij}^a is indirect and subleading, its effect on spectral isolation is direct and favorable.

K. Effect on mass protection

The dangerous constant masses on the minimal doubled Dirac block are

$$\rho^1 \otimes \mathbf{1}, \quad \rho^2 \otimes \mathbf{1}.$$

The stiffness term is derivative and projector-valued:

$$\mathcal{F}_{\text{align}} \propto \int \rho^2 (\partial P \partial P).$$

Therefore it does not itself introduce a constant intervalley mass at tree level.

After heavy-mode elimination, it may renormalize derivative couplings and remote-sector energies, but it does not naturally generate a constant mass unless the projector sector explicitly breaks the residual symmetry protecting valley charge.

So, again, the stiffness term is more naturally a stabilizer than a mass-generator.

L. Main theorem

[Role of the projector stiffness term] Consider the refined microscopic model with

$$\mathcal{F}_{\text{align}} = \frac{\chi}{4} \int d^3x \rho^2 (\partial_i P \partial_i P).$$

Expand around a locked constant-projector background $P_0 = \text{const}$.

Then:

1. $\mathcal{F}_{\text{align}}$ vanishes on the background and begins at quadratic order in projector fluctuations;
2. it does not generically generate a new tree-level enriched tensor N_{ij}^a on the locked background;

3. it gives projector tangent modes a positive inverse propagator of order

$$\chi \bar{\rho}^2 q_*^2;$$

4. integrating out these heavy modes can generate effective corrections

$$\delta N_{ij}^a, \quad \delta W_{\parallel}, \quad \delta W_I, \quad \delta W_J$$

of order

$$O\left(\frac{1}{\chi \bar{\rho}^2 q_*^2}\right)$$

provided the microscopic mixing source has the appropriate shell-gradient/test-derivative structure;

5. it directly enhances the remote-sector gap and tends to preserve opposite-valley matching.

Items (1) and (2) follow from the expansion around $P_0 = \text{const}$, for which $\partial_i P_0 = 0$. Item (3) follows from the quadratic tangent-mode operator. Item (4) follows from standard Gaussian elimination of the heavy projector mode. Item (5) follows because the projector tangent modes lie in the remote sector and the stiffness term is even under opposite-valley exchange.

M. What this means operationally

So the correct operational answer is:

Yes, the projector stiffness contribution should eventually be computed.

(109)

But the right computation is not

“write $\mathcal{F}_{\text{align}}$ and read off N_{ij}^a directly.”

The right computation is

expand around the locked background, identify the heavy projector mode, and integrate it out to obtain the induced shell-enrichment.

(110)

That is the mathematically correct route.

N. What is now closed and what remains open

What is now rigorously established:

1. the projector stiffness term is not irrelevant;
2. its direct tree-level effect on N_{ij}^a is generically zero on a locked constant-projector background;
3. its indirect effect after heavy-mode elimination can renormalize N_{ij}^a , the witness triple, and the Weyl determinant;
4. its most direct benefit is improved locking and an enhanced remote gap.

What remains open:

1. the explicit microscopic form of the mixing source $\mathcal{J}$ between shell modes and projector tangent modes;
2. the explicit induced correction δN_{ij}^a ;
3. the corresponding explicit corrections $\delta W_{\parallel}, \delta W_I, \delta W_J$ in the refined microscopic model.

XII. FINAL SUMMARY

The projector stiffness term

$$\mathcal{F}_{\text{align}} = \frac{\chi}{4} \int d^3x \rho^2 (\partial_i P \partial_i P)$$

should indeed be accounted for in the final microscopic TECT program.

However, on a locked constant-projector background $P_0 = \text{const}$, it does not generically generate a new tree-level shell-enriched tensor N_{ij}^a , because it vanishes on the background and starts only at quadratic order in projector fluctuations.

Its main role is instead to make projector tangent modes heavy, with characteristic inverse propagator scale

$$\chi \bar{\rho}^2 q_*^2.$$

After integrating out these heavy modes, one can obtain induced corrections

$$\delta N_{ij}^a, \quad \delta W_{\parallel}, \quad \delta W_I, \quad \delta W_J$$

of order

$$O\left(\frac{1}{\chi \bar{\rho}^2 q_*^2}\right),$$

provided the microscopic shell sector couples to the projector tangent mode with the appropriate tensor structure.

Thus the projector stiffness term matters, but mainly indirectly: it stabilizes locking, enhances the remote spectral gap, and renormalizes the shell-enriched Weyl/Dirac data after heavy-mode elimination.

XIII. QUADRATIC EXPANSION AROUND THE LOCKED BACKGROUND AND THE ABSENCE OF A MIXING SOURCE $\mathcal{J}$

A. Goal

The purpose of this section is to answer the question rigorously:

Does the projector stiffness term generate a shell-projector mixing source $\mathcal{J}$ around the locked background?

We will show that, for the presently adopted refined $U(3)$ -invariant triplet model, the answer is *no* at quadratic order.

More precisely:

Around a locked constant-projector background, the quadratic fluctuation operator is block diagonal between the scalar shell s (111)

Therefore no linear mixing source $\mathcal{J}$ exists at this order, and hence no induced correction

$$\delta N_{ij}^a$$

can be derived from projector-stiffness elimination within the current refined model alone.

B. Refined static free energy

We work with the refined static free energy

$$\mathcal{F}_{\text{ref}}[\Psi] = \mathcal{F}_{\text{triplet}}[\Psi] + \mathcal{F}_{\text{align}}[\Psi], \quad (112)$$

where

$$\mathcal{F}_{\text{triplet}}[\Psi] = \int d^3x \left[\frac{r}{2} \Psi^\dagger \Psi + \frac{Z}{2} \partial_i \Psi^\dagger \partial_i \Psi + \frac{Y}{2} (\nabla^2 \Psi)^\dagger (\nabla^2 \Psi) + \frac{g}{4} (\Psi^\dagger \Psi)^2 \right], \quad (113)$$

and

$$\mathcal{F}_{\text{align}}[\Psi] = \frac{\chi}{4} \int d^3x \rho^2 (\partial_i P \partial_i P). \quad (114)$$

Here

$$\Psi = \rho z, \quad z^\dagger z = 1, \quad P = z z^\dagger. \quad (115)$$

C. Locked background

Let the background be a locked shell condensate

$$\Psi_0(x) = \phi_0(x) z_0, \quad z_0^\dagger z_0 = 1, \quad z_0 = \text{const}, \quad (116)$$

where $\phi_0(x)$ is the scalar BCC shell background solving the scalar Euler–Lagrange equation in the locked sector.

Then

$$P_0 = z_0 z_0^\dagger = \text{const}, \quad \partial_i P_0 = 0. \quad (117)$$

D. Fluctuation decomposition

We now decompose fluctuations into:

1. a longitudinal shell fluctuation $\sigma(x)$ along z_0 ,
2. a tangent projector fluctuation $\xi(x) \in z_0^\perp \subset \mathbb{C}^3$.

Write

$$\Psi(x) = (\phi_0(x) + \sigma(x)) z_0 + \xi(x), \quad z_0^\dagger \xi(x) = 0. \quad (118)$$

This is the natural orthogonal decomposition of the triplet field around the locked background.

E. Quadratic expansion of the norm term

First compute

$$\begin{aligned} \Psi^\dagger \Psi &= (\phi_0 + \sigma)^* (\phi_0 + \sigma) z_0^\dagger z_0 + \xi^\dagger \xi + (\phi_0 + \sigma)^* z_0^\dagger \xi + (\phi_0 + \sigma) \xi^\dagger z_0 \\ &= |\phi_0 + \sigma|^2 + \xi^\dagger \xi, \end{aligned} \quad (119)$$

because

$$z_0^\dagger \xi = \xi^\dagger z_0 = 0.$$

Thus there is no term linear in both σ and ξ coming from $\Psi^\dagger \Psi$.

F. Quadratic expansion of the gradient term

Since z_0 is constant,

$$\partial_i \Psi = (\partial_i \phi_0 + \partial_i \sigma) z_0 + \partial_i \xi. \quad (120)$$

Hence

$$\partial_i \Psi^\dagger \partial_i \Psi = |\partial_i \phi_0 + \partial_i \sigma|^2 + \partial_i \xi^\dagger \partial_i \xi + (\partial_i \phi_0 + \partial_i \sigma)^* z_0^\dagger \partial_i \xi + (\partial_i \phi_0 + \partial_i \sigma) \partial_i \xi^\dagger z_0. \quad (121)$$

But

$$z_0^\dagger \partial_i \xi = \partial_i (z_0^\dagger \xi) - (\partial_i z_0^\dagger) \xi = 0,$$

because z_0 is constant and $z_0^\dagger \xi = 0$. Therefore the cross terms vanish, and

$$\partial_i \Psi^\dagger \partial_i \Psi = |\partial_i \phi_0 + \partial_i \sigma|^2 + \partial_i \xi^\dagger \partial_i \xi. \quad (122)$$

So again there is no quadratic mixing between σ and ξ .

G. Quadratic expansion of the biharmonic term

Similarly,

$$\nabla^2 \Psi = (\nabla^2 \phi_0 + \nabla^2 \sigma) z_0 + \nabla^2 \xi, \quad (123)$$

so

$$\begin{aligned} (\nabla^2 \Psi)^\dagger (\nabla^2 \Psi) &= |\nabla^2 \phi_0 + \nabla^2 \sigma|^2 + (\nabla^2 \xi)^\dagger (\nabla^2 \xi) \\ &\quad + (\nabla^2 \phi_0 + \nabla^2 \sigma)^* z_0^\dagger \nabla^2 \xi + (\nabla^2 \phi_0 + \nabla^2 \sigma) (\nabla^2 \xi)^\dagger z_0. \end{aligned} \quad (124)$$

Again

$$z_0^\dagger \nabla^2 \xi = \nabla^2 (z_0^\dagger \xi) = 0,$$

so

$$(\nabla^2 \Psi)^\dagger (\nabla^2 \Psi) = |\nabla^2 \phi_0 + \nabla^2 \sigma|^2 + (\nabla^2 \xi)^\dagger (\nabla^2 \xi). \quad (125)$$

Thus the Y -term also yields no quadratic shell-projector mixing.

H. Quadratic expansion of the quartic term

Using (119),

$$(\Psi^\dagger \Psi)^2 = (|\phi_0 + \sigma|^2 + \xi^\dagger \xi)^2.$$

Expanding to quadratic order in fluctuations gives

$$(\Psi^\dagger \Psi)^2 = \phi_0^4 + 4\phi_0^3 \Re \sigma + (\text{quadratic terms in } \sigma) + 2\phi_0^2 \xi^\dagger \xi + O(3). \quad (126)$$

There is no term bilinear in σ and ξ . Hence the quartic term also does not generate quadratic shell-projector mixing.

I. Quadratic expansion of the projector stiffness term

Now consider the alignment term. Around the locked background,

$$P = P_0 + \delta P + O((\delta P)^2), \quad \delta P = z_0 \xi^\dagger + \xi z_0^\dagger + O(\xi^2).$$

Since $\partial_i P_0 = 0$,

$$\partial_i P = \partial_i \delta P + O(\xi \partial \xi, \xi^2). \quad (127)$$

Therefore

$$(\partial_i P \partial_i P) = (\partial_i \delta P \partial_i \delta P) + O(3), \quad (128)$$

and

$$\mathcal{F}_{\text{align}}^{(2)} = \frac{\chi}{4} \int d^3 x \phi_0(x)^2 (\partial_i \delta P \partial_i \delta P). \quad (129)$$

This depends only on ξ to quadratic order. There is no bilinear σ - ξ term.

J. Quadratic block-diagonalization theorem

Collecting the previous expansions, we obtain:

[Quadratic block diagonalization around the locked background] Let

$$\Psi_0(x) = \phi_0(x)z_0$$

be a locked background with z_0 constant, and decompose fluctuations as

$$\Psi = (\phi_0 + \sigma)z_0 + \xi, \quad z_0^\dagger \xi = 0.$$

Then the quadratic fluctuation operator of the refined $U(3)$ -invariant triplet model is block diagonal:

$$\delta^2 \mathcal{F}_{\text{ref}} = \delta^2 \mathcal{F}_\sigma + \delta^2 \mathcal{F}_\xi, \quad (130)$$

with no bilinear σ - ξ term. Equivalently, the shell fluctuation sector and the tangent projector sector do not mix at quadratic order.

The absence of bilinear σ - ξ terms was shown separately for the norm term, gradient term, biharmonic term, quartic term, and projector-stiffness term. Summing them gives the stated block-diagonal quadratic Hessian.

K. Consequences for the mixing source $\mathcal{J}$

In the previous discussion, a correction

$$\delta N_{ij}^a$$

from integrating out projector fluctuations would require a bilinear mixing source of schematic form

$$\mathcal{L}_{\text{mix}} = \mathcal{J}[\sigma, \psi] \cdot \xi, \quad (131)$$

where ψ denotes the shell-reduced test mode sector.

But Theorem 1 shows that, within the current refined $U(3)$ -invariant model and around the locked constant-projector background, there is no such quadratic source already at the level of the (σ, ξ) Hessian.

Therefore:

$$\boxed{\mathcal{J} = 0 \quad \text{at quadratic order in the current refined model.}} \quad (132)$$

This implies that no induced correction

$$\delta N_{ij}^a$$

arises from simply integrating out the projector tangent mode in the present model at this order.

L. Consequences for N_{ij}^a , the witness triple, and the Weyl/Dirac criteria

Because $\mathcal{J} = 0$ at quadratic order:

1. the projector stiffness term does not generate a new effective shell-enriched tensor N_{ij}^a at this order;
2. therefore it does not modify the witness triple

$$(W_{\parallel}, W_I, W_J)$$

through a quadratic induced term at this order;

3. hence it does not directly alter the leading Weyl determinant

$$\det M = (\kappa_* |G_*|)^3 W_{\parallel} (W_I^2 + W_J^2)$$

through such an induced correction.

Its role remains:

- improving locking,
- lifting projector tangent modes into the remote sector,
- and enhancing spectral isolation.

M. Interpretation

This calculation shows that the current refined model is still not enough to produce the full shell-enriched TECT Dirac structure dynamically.

The projector stiffness helps by stabilizing the locked background and by increasing the remote gap, but it does not by itself generate the missing gradient-linear enriched tensor sector.

Thus the logical conclusion is:

If one wants nonzero induced δN_{ij}^a , one must add an extra UV coupling that is linear in the tangent projector fluctuation and
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(133)

N. Minimal structural requirement for a nonzero $\mathcal{J}$

The missing UV sector must supply a coupling whose quadratic expansion around the locked background contains a term of the schematic form

$$\mathcal{L}_{\text{UV,mix}}^{(2)} \sim (\partial_i \phi_0) (\text{test derivative/internal factor}) \xi. \quad (134)$$

Only after such a term exists can one integrate out ξ and obtain a genuine induced correction

$$\delta N_{ij}^a.$$

So the correct next inverse problem is not “compute δN from the current model,” but rather:

find the minimal extra UV coupling that generates a nonzero quadratic mixing source $\mathcal{J}$.

(135)

O. What is now closed and what remains open

What is now rigorously established:

1. in the current refined $U(3)$ -invariant model, the locked-background quadratic Hessian is block diagonal between shell and tangent-projector sectors;
2. therefore

$$\mathcal{J} = 0$$

at quadratic order;

3. hence projector stiffness does not generate δN_{ij}^a , δW_a , or a correction to the Weyl/Dirac criteria at this order;
4. its direct mathematical effect is instead improved locking and enhanced remote spectral separation.

What remains open:

1. the identification of the minimal extra UV interaction that yields a nonzero quadratic shell–projector mixing source;
2. the induced shell-enriched tensor δN_{ij}^a once such a UV interaction is specified;
3. the corresponding corrected witness triple and Weyl/Dirac data in the fully extended microscopic model.

XIV. FINAL SUMMARY

For the current refined $U(3)$ -invariant triplet model

$$\mathcal{F}_{\text{ref}} = \mathcal{F}_{\text{triplet}} + \frac{\chi}{4} \int d^3x \rho^2 \text{Tr}(\partial_i P \partial_i P),$$

expand around the locked background

$$\Psi_0(x) = \phi_0(x)z_0, \quad z_0 = \text{const},$$

and decompose fluctuations as

$$\Psi = (\phi_0 + \sigma)z_0 + \xi, \quad z_0^\dagger \xi = 0.$$

Then the quadratic fluctuation Hessian is block diagonal:

$$\delta^2 \mathcal{F}_{\text{ref}} = \delta^2 \mathcal{F}_\sigma + \delta^2 \mathcal{F}_\xi,$$

with no bilinear σ - ξ term.

Therefore the shell sector and the tangent projector sector do not mix at quadratic order, so the corresponding mixing source vanishes:

$$\mathcal{J} = 0.$$

Consequently, within the current refined model, the projector stiffness term does not generate an induced correction

$$\delta N_{ij}^a$$

at quadratic order, and therefore does not directly modify the leading witness triple

$$(W_{\parallel}, W_I, W_J)$$

or the leading Weyl/Dirac criteria at this order.

Its actual direct role is instead to stabilize the locked sector and to push projector tangent modes upward into the remote sector.

Thus, if one wants a genuine induced shell-enriched tensor correction, one must introduce an additional UV coupling whose quadratic expansion contains a nonzero shell-projector mixing source.

XV. MINIMAL UV EXTENSIONS THAT CAN GENERATE A NONZERO QUADRATIC MIXING SOURCE $\mathcal{J}$

A. Goal

The current refined $U(3)$ -invariant triplet model

$$\mathcal{F}_{\text{ref}} = \mathcal{F}_{\text{triplet}} + \mathcal{F}_{\text{align}}$$

was shown to have a block-diagonal quadratic Hessian around the locked constant-projector background. Therefore

$$\mathcal{J} = 0$$

at quadratic order in that model.

The purpose of this section is to classify the minimal UV extensions that can change this conclusion.

More precisely, we seek the smallest additional microscopic structures that can generate, around the locked background,

$$\mathcal{L}_{\text{mix}}^{(2)} \sim \mathcal{J}[\sigma, \psi] \cdot \xi, \tag{136}$$

where:

- σ denotes the shell/amplitude fluctuation sector,
- ψ denotes the shell-reduced test/fermionic/internal mode sector,
- ξ denotes the tangent projector fluctuation.

Only after such a nonzero quadratic mixing source exists can integrating out ξ induce a genuine correction

$$\delta N_{ij}^a.$$

B. Structural requirements on a nonzero mixing source

We first determine what any admissible $\mathcal{J}$ must look like.

The shell-enriched tensor we ultimately want has the schematic form

$$\delta\mathcal{L}_{\text{enr}} = N_{ij}^a (\partial_i \phi_0) (-i\partial_j) \otimes s_a. \quad (137)$$

Therefore any UV source that can induce N_{ij}^a through projector-mode elimination must, at quadratic order around the locked background, satisfy all of the following:

1. it must be linear in the tangent projector fluctuation ξ ;
2. it must be linear in the shell/background gradient $\partial_i \phi_0$ or an equivalent shell-direction tensor;
3. it must be linear in a derivative acting on the test mode sector, so that after elimination one obtains a first-order enriched operator;
4. it must carry an internal triplet label a matching the reduced shell Pauli sector s_a .

Hence the generic target form is

$$\mathcal{L}_{\text{mix}}^{(2)} = \Lambda_{ij}^a (\partial_i \phi_0) (\mathcal{D}_j \psi) \xi_a + \text{h.c.}, \quad (138)$$

or an equivalent mediated version thereof.

This already implies that the UV extension must introduce an object carrying both a spatial index i (or j) and an internal index a .

C. Why purely scalar or purely projector corrections are insufficient

Before classifying viable UV extensions, it is useful to rule out insufficient ones.

a. Type 0A: additional scalar potential terms. Terms such as

$$(\Psi^\dagger \Psi)^3, \quad (\Psi^\dagger \Psi)(\partial_i P \partial_i P), \quad ((\partial_i P \partial_i P))^2$$

do not generate a linear shell–projector mixing source around a locked constant-projector background. They are either polarization-blind or start at quadratic order in ξ without a term linear in ξ .

b. Type 0B: additional pure projector derivative terms. Terms such as

$$(\partial_i P \partial_i P), \quad (\nabla^2 P \nabla^2 P), \quad ((\partial_i P \partial_i P))^2$$

also fail for the same reason: they are purely projector-sector terms and yield no linear source coupling ξ to the shell/test sector.

c. Type 0C: additional amplitude-only derivative terms. Terms depending only on ρ and its derivatives cannot introduce the required internal label a .

Thus:

To obtain $\mathcal{J} \neq 0$, the UV extension must explicitly couple the shell derivative sector to an internal tangent/projector carrier.
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D. Classification principle

We now classify the minimal UV extensions by the nature of the new field or tensor that mediates the coupling between:

1. the shell/background derivative structure,
2. the internal tangent/projector sector,
3. the test-mode derivative/internal channel.

At lowest derivative order, there are three natural classes.

E. Class I: direct shell–projector derivative couplings

The first possibility is to add a direct local term involving Ψ , P , and one derivative on each side. A minimal representative is

$$\mathcal{L}_{\text{UV}}^{(\text{I})} = \Lambda_{ij}^a (\partial_i \Psi^\dagger) T_a \Psi (T_a \partial_j P) + \text{h.c.}, \quad (140)$$

where T_a are Hermitian generators adapted to the shell-relevant internal channels.

a. Why this can work. Expand around the locked background

$$\Psi = (\phi_0 + \sigma)z_0 + \psi, \quad P = P_0 + \delta P, \quad \delta P \sim z_0 \xi^\dagger + \xi z_0^\dagger.$$

Then

$$\partial_j P \sim \partial_j \delta P \sim \partial_j \xi,$$

while

$$\partial_i \Psi^\dagger T_a \Psi$$

contains terms linear in $\partial_i \phi_0$ and linear in the test/internal fluctuation sector. Therefore (140) can generate at quadratic order a source of the form

$$(\partial_i \phi_0)(\partial_j \psi) \xi_a.$$

b. Caveat. This class is algebraically minimal, but somewhat ad hoc: it directly inserts the desired tensor structure into the UV Lagrangian. So it is useful as a mathematical existence proof, but less compelling as a fundamental physical origin.

F. Class II: auxiliary internal vector mediator

The second, and structurally cleaner, possibility is to introduce an auxiliary field

$$X_i^a(x), \quad (141)$$

carrying one spatial index and one internal adjoint/reduced-shell index.

A minimal mediated UV Lagrangian is

$$\mathcal{L}_{\text{UV}}^{(\text{II})} = \frac{M_X^2}{2} X_i^a X_i^a + \alpha X_i^a J_i^a[\Psi] + \beta X_i^a K_i^a[P, \psi]. \quad (142)$$

Here:

- $J_i^a[\Psi]$ is a shell/background current built from Ψ ,
- $K_i^a[P, \psi]$ is a projector/test-mode current built from P and ψ .

The minimal choices are:

$$J_i^a[\Psi] = (\partial_i \Psi^\dagger) T_a \Psi + \text{h.c.}, \quad (143)$$

$$K_i^a[P, \psi] = (T_a \partial_i P) \mathcal{O}[\psi], \quad (144)$$

or, after shell reduction,

$$K_i^a \sim \partial_i \delta P^a \psi.$$

Integrating out the auxiliary field X_i^a gives

$$\mathcal{L}_{\text{UV,eff}}^{(\text{II})} = -\frac{\alpha\beta}{M_X^2} J_i^a[\Psi] K_i^a[P, \psi] + \dots, \quad (145)$$

which produces the desired quadratic shell–projector source.

a. Why this class is especially natural. This class is preferable because:

1. the mediator carries exactly the required tensor structure (i, a) ;
2. the mixing is generated dynamically by elimination rather than inserted directly;
3. the size of the induced coupling is calculable:

$$\delta N \sim \frac{\alpha\beta}{M_X^2};$$

4. by making M_X large one can control the derivative expansion consistently.

Thus Class II is the most natural minimal UV extension for generating $\mathcal{J} \neq 0$.

G. Class III: auxiliary connection / gauge-like mediator

The third possibility is to introduce a connection-like field

$$A_i^a(x), \tag{146}$$

coupled both to the internal projective sector and to the shell/test sector.

A minimal representative is

$$\mathcal{L}_{\text{UV}}^{(\text{III})} = \frac{M_A^2}{2} A_i^a A_i^a + \gamma A_i^a (T_a \partial_i P) + \delta A_i^a \mathcal{J}_i^a[\Psi, \psi]. \tag{147}$$

Integrating out A_i^a gives

$$\mathcal{L}_{\text{UV,eff}}^{(\text{III})} = -\frac{\gamma\delta}{M_A^2} (T_a \partial_i P) \mathcal{J}_i^a[\Psi, \psi] + \dots \tag{148}$$

Again, after expansion around the locked background, this can generate the desired shell-projector quadratic source.

a. Interpretation. This class is physically appealing if one wants the internal projective geometry to come from an emergent connection/gauge sector. It is less minimal than Class II algebraically, but potentially deeper conceptually.

H. Minimality comparison

We now compare the three classes.

a. Class I (direct coupling). Smallest in terms of field content, but most ad hoc: it directly inserts the mixed tensor operator.

b. Class II (auxiliary internal vector mediator). Smallest mediated completion. It introduces only one heavy auxiliary field X_i^a with exactly the right index structure and generates the mixing dynamically.

c. Class III (connection-like mediator). Most conceptually rich, especially if one expects emergent gauge structure, but slightly less minimal.

Therefore the most natural minimal UV extension is:

Class II: a heavy auxiliary internal vector mediator X_i^a .

(149)

I. Explicit quadratic source from Class II

We now show explicitly how the source appears.

Take

$$\mathcal{L}_{\text{UV}}^{(\text{II})} = \frac{M_X^2}{2} X_i^a X_i^a + \alpha X_i^a (\partial_i \Psi^\dagger T_a \Psi + \text{h.c.}) + \beta X_i^a \partial_i \delta P^a \psi. \tag{150}$$

Around the locked background

$$\Psi_0 = \phi_0 z_0, \quad \delta P^a \sim \xi^a,$$

the shell current contains the piece

$$\partial_i \Psi^\dagger T_a \Psi = (\partial_i \phi_0) z_0^\dagger T_a z_0 \phi_0 + (\partial_i \phi_0) \mathcal{C}_a[\psi] + \dots \quad (151)$$

The second term is the relevant shell/test-mode source. Eliminating X_i^a produces

$$\mathcal{L}_{\text{mix}}^{(2)} \sim -\frac{\alpha\beta}{M_X^2} (\partial_i \phi_0) \mathcal{C}_a[\psi] \partial_i \xi^a. \quad (152)$$

After integration by parts / shell reduction, this is precisely of the target form

$$(\partial_i \phi_0) (\partial_j \psi) \xi^a,$$

hence a nonzero $\mathcal{J}$.

Consequently, integrating out ξ can now genuinely generate

$$\delta N_{ij}^a \neq 0.$$

J. Induced scale of δN_{ij}^a

If the tangent projector mode has inverse propagator

$$\mathcal{G}_\xi^{-1}(k) \sim \chi_{\text{eff}} k^2 + m_\xi^2,$$

then eliminating ξ after generating the source (152) yields schematically

$$\delta N_{ij}^a \sim \frac{\alpha\beta}{M_X^2} \frac{1}{\chi_{\text{eff}} q_*^2 + m_\xi^2} \times (\text{projected tensor coefficient}). \quad (153)$$

This is the first mathematically consistent route to a derived nonzero shell-enriched tensor correction from projector-sector physics.

K. Compatibility with the shell/Weyl/Dirac framework

Any acceptable UV extension must not destroy the previously derived structure. The three viable classes above preserve the framework if:

1. they vanish or reduce to controlled perturbations on the locked background;
2. they do not create a dominant constant intervalley mass term;
3. they preserve the shell-relevant opposite-valley symmetry to leading order;
4. their induced corrections remain subleading relative to the locked leading shell witness structure.

Class II can satisfy all of these naturally by choosing the mediator heavy and the couplings sufficiently small compared to the shell-scale leading terms.

L. Main classification theorem

[Minimal UV classification for nonzero quadratic shell-projector mixing] Assume one wants a nonzero quadratic mixing source

$$\mathcal{L}_{\text{mix}}^{(2)} \sim \mathcal{J}[\sigma, \psi] \cdot \xi$$

around the locked constant-projector background, while preserving locality and the shell-relevant derivative structure needed for an induced correction δN_{ij}^a .

Then any minimal UV extension must introduce, explicitly or effectively, an object carrying both:

1. one spatial derivative index,
2. one internal reduced-shell index a .

At lowest derivative order, the minimal viable extensions fall into three classes:

1. direct shell–projector derivative couplings;
2. heavy auxiliary internal vector mediators X_i^a ;
3. heavy connection/gauge-like mediators A_i^a .

Among these, the structurally cleanest minimal mediated completion is Class II, based on a heavy auxiliary internal vector X_i^a .

The necessity of an (i, a) -carrier follows from the target enriched tensor structure. Pure scalar or pure projector additions fail to generate the required quadratic source. The three listed classes exhaust the lowest-order local possibilities for supplying the missing tensor carrier either directly or through mediation. Class II is the minimal mediated realization.

M. What is now closed and what remains open

What is now rigorously established:

1. the current refined model is insufficient because it has $\mathcal{J} = 0$;
2. any nonzero quadratic shell–projector mixing requires an extra UV structure carrying a spatial and an internal index;
3. the minimal viable UV extensions are classified into three classes;
4. the most natural minimal mediated class is the heavy auxiliary internal vector X_i^a .

What remains open:

1. which of the three classes is actually realized in the true microscopic TECT;
2. the explicit shell-reduced induced tensor δN_{ij}^a for the chosen UV completion;
3. the corresponding corrected witness triple and Weyl/Dirac data.

XVI. FINAL SUMMARY

To generate a nonzero quadratic shell–projector mixing source

$$\mathcal{L}_{\text{mix}}^{(2)} \sim \mathcal{J}[\sigma, \psi] \cdot \xi$$

around the locked background, the microscopic theory must contain more than the current refined $U(3)$ -invariant triplet model.

The missing UV structure must carry both:

$$\text{one spatial derivative index } i, \quad \text{one internal reduced-shell index } a.$$

Pure scalar or pure projector corrections are insufficient.

At lowest derivative order, the minimal viable UV extensions fall into three classes:

1. direct shell–projector derivative couplings;
2. heavy auxiliary internal vector mediators X_i^a ;
3. heavy connection/gauge-like mediators A_i^a .

Among these, the most natural minimal mediated completion is a heavy auxiliary internal vector field X_i^a , because it dynamically generates the desired mixing after elimination and gives a calculable induced correction

$$\delta N_{ij}^a \sim \frac{\alpha\beta}{M_X^2} \frac{1}{\chi_{\text{eff}} q_*^2 + m_\xi^2} \times (\text{projected tensor coefficient}).$$

Thus the next true microscopic step is not more shell-level algebra, but choosing the minimal UV completion class and computing the induced shell-enriched tensor from it.

XVII. EXPLICIT COMPUTATION OF THE INDUCED TENSOR δN_{ij}^a IN THE MINIMAL CLASS II UV EXTENSION

A. Goal

We now choose one concrete representative of Class II and compute the induced shell-enriched tensor δN_{ij}^a explicitly.

The purpose is not yet to claim that this is the unique true UV completion of TECT. The purpose is to show, in a mathematically controlled minimal mediated model, how a nonzero shell-enriched tensor correction is actually generated.

We shall prove that a heavy auxiliary internal vector field X_i^a produces an effective first-order shell-enriched operator of the form

$$\delta \mathcal{L}_{\text{eff}} = \delta N_{ij}^a (\partial_i \phi_0) (-i \partial_j) s_a \psi,$$

with

$$\boxed{\delta N_{ij}^b = -\frac{\lambda g_X}{M_X^2} m^a C_{ij}^{ab}.} \quad (154)$$

This is the first explicit nonzero microscopic route to the enriched tensor correction.

B. Locked background data

Let the locked shell-condensed background be

$$\Psi_0(x) = \phi_0(x) z_0, \quad z_0^\dagger z_0 = 1, \quad P_0 = z_0 z_0^\dagger. \quad (155)$$

Assume the shell profile $\phi_0(x)$ is one of the scalar shell-condensed backgrounds (for example the BCC shell background).

The reduced shell test mode is denoted by ψ , and its internal reduced-shell Pauli channels are denoted by s_a , $a = 1, 2, 3$.

C. Minimal Class II UV extension

Introduce a heavy auxiliary internal vector mediator

$$X_i^a(x), \quad i = 1, 2, 3, \quad a = 1, 2, 3, \quad (156)$$

with quadratic UV Lagrangian

$$\mathcal{L}_X = \frac{M_X^2}{2} X_i^a X_i^a. \quad (157)$$

We now couple X_i^a to two currents:

1. a background shell/projector current $J_i^a[\phi_0, P_0]$,
2. a shell-reduced test-mode derivative current $K_i^a[\psi]$.

The minimal quadratic Class II extension is

$$\boxed{\mathcal{L}_{\text{UV}}^{(2)} = \frac{M_X^2}{2} X_i^a X_i^a + \lambda X_i^a J_i^a[\phi_0, P_0] + g_X X_i^a K_i^a[\psi].} \quad (158)$$

D. Choice of the two currents

We now choose the minimal currents consistent with the target shell-enriched structure.

a. Background current. The background current must be linear in the shell gradient and must carry an internal index a . The most general lowest-order local form is

$$J_i^a[\phi_0, P_0] = m^a(P_0) \partial_i \phi_0, \quad (159)$$

where

$$m^a(P_0) \quad (160)$$

is a constant internal vector determined by the locked background projector P_0 .

A canonical example is

$$m^a(P_0) = (T^a P_0), \quad (161)$$

for some chosen Hermitian generator basis T^a , but we do not need this explicit realization in what follows. Only the existence of a nonzero constant vector m^a matters.

b. Test derivative current. The current coupled to the test mode must be linear in a derivative acting on ψ , and must carry the same mediator index a . The most general lowest-order linear form is

$$K_i^a[\psi] = \mathcal{C}_{ij}^{ab} (-i\partial_j) s_b \psi, \quad (162)$$

where $\mathcal{C}_{ij}^{ab}$ is a fixed UV tensor kernel supplied by the auxiliary mediator sector.

Thus the concrete Class II quadratic extension becomes

$$\mathcal{L}_{\text{UV}}^{(2)} = \frac{M_X^2}{2} X_i^a X_i^a + \lambda X_i^a m^a \partial_i \phi_0 + g_X X_i^a \mathcal{C}_{ij}^{ab} (-i\partial_j) s_b \psi. \quad (163)$$

E. Integrating out the heavy auxiliary mediator

Because X_i^a is auxiliary and heavy, its equation of motion is algebraic:

$$\frac{\partial \mathcal{L}_{\text{UV}}^{(2)}}{\partial X_i^a} = 0 \quad \implies \quad M_X^2 X_i^a + \lambda m^a \partial_i \phi_0 + g_X \mathcal{C}_{ij}^{ab} (-i\partial_j) s_b \psi = 0. \quad (164)$$

Therefore

$$X_i^a = -\frac{1}{M_X^2} [\lambda m^a \partial_i \phi_0 + g_X \mathcal{C}_{ij}^{ab} (-i\partial_j) s_b \psi]. \quad (165)$$

Substituting (165) back into (163), we obtain the effective quadratic contribution

$$\delta \mathcal{L}_{\text{eff}} = -\frac{1}{2M_X^2} [\lambda m^a \partial_i \phi_0 + g_X \mathcal{C}_{ij}^{ab} (-i\partial_j) s_b \psi] [\lambda m^a \partial_i \phi_0 + g_X \mathcal{C}_{ik}^{ac} (-i\partial_k) s_c \psi]. \quad (166)$$

This contains three types of terms:

1. a pure background term $\sim (\partial \phi_0)^2$,
2. a pure test-mode derivative term $\sim ((-i\partial)\psi)^2$,
3. the desired cross term linear in $\partial_i \phi_0$ and linear in $(-i\partial_j) s_b \psi$.

The cross term is

$$\delta \mathcal{L}_{\text{cross}} = -\frac{\lambda g_X}{M_X^2} m^a \mathcal{C}_{ij}^{ab} (\partial_i \phi_0) (-i\partial_j) s_b \psi. \quad (167)$$

Comparing (167) with the target shell-enriched form

$$\delta\mathcal{L}_{\text{enr}} = \delta N_{ij}^b (\partial_i \phi_0) (-i\partial_j) s_b \psi,$$

we read off immediately

$$\boxed{\delta N_{ij}^b = -\frac{\lambda g_X}{M_X^2} m^a C_{ij}^{ab}.} \quad (168)$$

This proves the main formula.

F. Shell-frame decomposition of the induced tensor

To connect δN_{ij}^a with the witness triple, decompose the induced tensor in the shell-adapted basis

$$\{\hat{n}, e_1, e_2\}.$$

Write the shell-relevant projected part of $m^a C_{ij}^{ab}$ as

$$\begin{aligned} m^a C_{ij}^{ab} &= A \hat{n}_i \hat{n}_j \delta^{b3} \\ &+ B (\hat{n}_i e_{1j} \delta^{b1} + \hat{n}_i e_{2j} \delta^{b2}) \\ &+ C (-\hat{n}_i e_{2j} \delta^{b1} + \hat{n}_i e_{1j} \delta^{b2}) \\ &+ (\text{components orthogonal to the shell witness projections}). \end{aligned} \quad (169)$$

Then the induced shell-enriched tensor becomes

$$\delta N_{ij}^3 = -\frac{\lambda g_X}{M_X^2} A \hat{n}_i \hat{n}_j + \dots, \quad (170)$$

$$\delta N_{ij}^1 = -\frac{\lambda g_X}{M_X^2} (B \hat{n}_i e_{1j} - C \hat{n}_i e_{2j}) + \dots, \quad (171)$$

$$\delta N_{ij}^2 = -\frac{\lambda g_X}{M_X^2} (B \hat{n}_i e_{2j} + C \hat{n}_i e_{1j}) + \dots. \quad (172)$$

Here the ellipses denote tensor components annihilated by the shell witness contractions.

G. Induced corrections to the witness triple

Recall the witness functionals

$$W_{\parallel}[N] = \hat{n}_i \hat{n}_j N_{ij}^3, \quad (173)$$

$$W_I[N] = \frac{1}{2} (e_{1j} \hat{n}_i N_{ij}^1 + e_{2j} \hat{n}_i N_{ij}^2), \quad (174)$$

$$W_J[N] = \frac{1}{2} (e_{1j} \hat{n}_i N_{ij}^2 - e_{2j} \hat{n}_i N_{ij}^1). \quad (175)$$

Applying these to (170)–(172), we obtain

$$\boxed{\delta W_{\parallel} = -\frac{\lambda g_X}{M_X^2} A,} \quad (176)$$

$$\boxed{\delta W_I = -\frac{\lambda g_X}{M_X^2} B,} \quad (177)$$

$$\boxed{\delta W_J = -\frac{\lambda g_X}{M_X^2} C.} \quad (178)$$

Thus the auxiliary mediator directly renormalizes the three shell witnesses.

H. Induced correction to the Weyl determinant

Let the original locked-sector witnesses be

$$W_{\parallel}^{(0)}, \quad W_I^{(0)}, \quad W_J^{(0)}.$$

After including the Class II UV extension,

$$W_{\parallel} = W_{\parallel}^{(0)} + \delta W_{\parallel}, \quad W_I = W_I^{(0)} + \delta W_I, \quad W_J = W_J^{(0)} + \delta W_J. \quad (179)$$

The corrected Weyl determinant is

$$\det M = (\kappa_* |G_*|)^3 W_{\parallel} (W_I^2 + W_J^2). \quad (180)$$

Expanding to first order in the induced correction,

$$\begin{aligned} \delta(\det M) = (\kappa_* |G_*|)^3 & \left[\delta W_{\parallel} ((W_I^{(0)})^2 + (W_J^{(0)})^2) \right. \\ & \left. + 2W_{\parallel}^{(0)} (W_I^{(0)} \delta W_I + W_J^{(0)} \delta W_J) \right] + O((\delta W)^2). \end{aligned} \quad (181)$$

Therefore the auxiliary mediator can either:

1. renormalize an already-nonzero Weyl structure, or
2. generate a nonzero witness channel if one of the bare coefficients happened to vanish.

I. Opposite-valley pairing

If the background current vector $m^a(P_0)$ and the UV tensor kernel C_{ij}^{ab} are even under opposite-valley exchange, then the same computation at $-G_*$ gives the same coefficients A, B, C . Hence

$$\delta W_{\parallel}^- = \delta W_{\parallel}^+, \quad \delta W_I^- = \delta W_I^+, \quad \delta W_J^- = \delta W_J^+, \quad (182)$$

so opposite-valley witness matching survives.

Thus the Class II UV extension can generate nonzero δN_{ij}^a without spoiling Dirac pairing, provided its background current respects the same opposite-valley symmetry.

J. Scale and interpretation

The induced tensor scale is controlled by

$$\delta N \sim \frac{\lambda g_X}{M_X^2}. \quad (183)$$

This is exactly what one should expect from integrating out a heavy auxiliary mediator at tree level:

- stronger couplings λ, g_X enhance the enriched tensor,
- heavier mediator mass M_X suppresses it.

Unlike the projector-stiffness term alone, this mechanism generates a nonzero shell-enriched tensor correction already at tree level.

K. Main theorem

[Explicit induced shell-enriched tensor from the minimal Class II UV mediator] Consider the heavy auxiliary internal vector field X_i^a with quadratic UV coupling

$$\mathcal{L}_{\text{UV}}^{(2)} = \frac{M_X^2}{2} X_i^a X_i^a + \lambda X_i^a m^a \partial_i \phi_0 + g_X X_i^a \mathcal{C}_{ij}^{ab} (-i\partial_j) s_b \psi.$$

Then integrating out X_i^a produces the effective shell-enriched operator

$$\delta\mathcal{L}_{\text{eff}} = \delta N_{ij}^b (\partial_i \phi_0) (-i\partial_j) s_b \psi,$$

with

$$\delta N_{ij}^b = -\frac{\lambda g_X}{M_X^2} m^a \mathcal{C}_{ij}^{ab}.$$

Its shell witness corrections are

$$\delta W_{\parallel} = -\frac{\lambda g_X}{M_X^2} A, \quad \delta W_I = -\frac{\lambda g_X}{M_X^2} B, \quad \delta W_J = -\frac{\lambda g_X}{M_X^2} C,$$

where A, B, C are the shell-adapted coefficients of the projected tensor $m^a \mathcal{C}_{ij}^{ab}$.

Solve the auxiliary field algebraically, substitute back, and compare the cross term with the standard shell-enriched operator form. The witness corrections then follow by direct contraction with the shell-adapted basis.

L. What is now closed and what remains open

What is now rigorously established:

1. a concrete minimal Class II UV extension has been chosen;
2. its heavy auxiliary mediator generates a nonzero shell-enriched tensor correction at tree level;
3. the induced tensor δN_{ij}^a is explicit;
4. the induced witness corrections $\delta W_{\parallel}, \delta W_I, \delta W_J$ are explicit;
5. opposite-valley pairing is preserved if the UV background current is even under valley exchange.

What remains open:

1. how $m^a(P_0)$ and $\mathcal{C}_{ij}^{ab}$ should be derived from a deeper UV TECT completion rather than inserted as the minimal Class II data;
2. whether the true microscopic TECT chooses this Class II route or another class in the final UV completion;
3. the full combined BCC/Weyl/Dirac/spectral analysis after this induced tensor is added.

XVIII. FINAL SUMMARY

For the minimal Class II UV completion, introduce a heavy auxiliary internal vector mediator

$$X_i^a$$

with quadratic UV coupling

$$\mathcal{L}_{\text{UV}}^{(2)} = \frac{M_X^2}{2} X_i^a X_i^a + \lambda X_i^a m^a \partial_i \phi_0 + g_X X_i^a \mathcal{C}_{ij}^{ab} (-i\partial_j) s_b \psi.$$

Integrating out X_i^a yields the effective shell-enriched correction

$$\delta\mathcal{L}_{\text{eff}} = \delta N_{ij}^b (\partial_i \phi_0) (-i\partial_j) s_b \psi,$$

with explicit induced tensor

$$\boxed{\delta N_{ij}^b = -\frac{\lambda g_X}{M_X^2} m^a C_{ij}^{ab}.$$

After decomposition into the shell-adapted basis, the induced witness corrections are

$$\delta W_{\parallel} = -\frac{\lambda g_X}{M_X^2} A, \quad \delta W_I = -\frac{\lambda g_X}{M_X^2} B, \quad \delta W_J = -\frac{\lambda g_X}{M_X^2} C.$$

Thus the heavy auxiliary internal vector mediator provides the first explicit microscopic mechanism that generates a nonzero shell-enriched tensor correction at tree level.

XIX. INSERTION OF THE CLASS II INDUCED TENSOR INTO THE EXISTING MICROSCOPIC MATCHING THEOREM

A. Short statement of the task

The attached TECT manuscript has already reduced the local Weyl problem to the three microscopic matching formulas

$$\begin{aligned} \lambda_{\parallel} &= \kappa_* |G_*| \hat{n}_i \hat{n}_j N_{ij}^3, \\ \alpha &= \frac{\kappa_* |G_*|}{2} \left(e_{1j} \hat{n}_i N_{ij}^1 + e_{2j} \hat{n}_i N_{ij}^2 \right), \\ \beta &= \frac{\kappa_* |G_*|}{2} \left(e_{2j} \hat{n}_i N_{ij}^1 - e_{1j} \hat{n}_i N_{ij}^2 \right), \end{aligned}$$

together with

$$\det M = \lambda_{\parallel} (\alpha^2 + \beta^2).$$

Therefore the correct next step is to insert the induced Class II correction

$$N_{ij}^a = N_{ij}^{a(0)} + \delta N_{ij}^a$$

into these already-closed formulas and derive the corrected Weyl data explicitly.

B. Baseline microscopic tensor and its shell-adapted coefficients

In the shell-adapted invariant basis, the previously established minimal microscopic tensor is

$$N_{ij}^{3(0)} = g_{\parallel} \hat{n}_i \hat{n}_j, \tag{184}$$

$$N_{ij}^{1(0)} = g_I \hat{n}_i e_{1j} - g_J \hat{n}_i e_{2j}, \tag{185}$$

$$N_{ij}^{2(0)} = g_I \hat{n}_i e_{2j} + g_J \hat{n}_i e_{1j}. \tag{186}$$

Hence the uncorrected matching data are

$$\lambda_{\parallel}^{(0)} = \kappa_* |G_*| g_{\parallel}, \quad \alpha^{(0)} = \kappa_* |G_*| g_I, \quad \beta^{(0)} = -\kappa_* |G_*| g_J. \tag{187}$$

Therefore

$$\det M^{(0)} = (\kappa_* |G_*|)^3 g_{\parallel} (g_I^2 + g_J^2). \tag{188}$$

C. Class II induced tensor

From the minimal heavy auxiliary internal vector mediator X_i^a , the induced shell-enriched correction was

$$\delta N_{ij}^b = -\frac{\lambda g_X}{M_X^2} m^a \mathcal{C}_{ij}^{ab}. \quad (189)$$

To connect to the matching theorem, decompose the shell-relevant part of the tensor

$$m^a \mathcal{C}_{ij}^{ab}$$

into the same invariant basis used above:

$$m^a \mathcal{C}_{ij}^{a3} = A \hat{n}_i \hat{n}_j + R_{ij}^{(3)}, \quad (190)$$

$$m^a \mathcal{C}_{ij}^{a1} = B \hat{n}_i e_{1j} - C \hat{n}_i e_{2j} + R_{ij}^{(1)}, \quad (191)$$

$$m^a \mathcal{C}_{ij}^{a2} = B \hat{n}_i e_{2j} + C \hat{n}_i e_{1j} + R_{ij}^{(2)}, \quad (192)$$

where $R_{ij}^{(a)}$ denotes the remainder orthogonal to the shell witness contractions, namely

$$\hat{n}_i \hat{n}_j R_{ij}^{(3)} = 0, \quad e_{1j} \hat{n}_i R_{ij}^{(1)} + e_{2j} \hat{n}_i R_{ij}^{(2)} = 0, \quad e_{2j} \hat{n}_i R_{ij}^{(1)} - e_{1j} \hat{n}_i R_{ij}^{(2)} = 0. \quad (193)$$

Then the shell-relevant induced tensor becomes

$$\delta N_{ij}^3 = -\frac{\lambda g_X}{M_X^2} A \hat{n}_i \hat{n}_j + \dots, \quad (194)$$

$$\delta N_{ij}^1 = -\frac{\lambda g_X}{M_X^2} (B \hat{n}_i e_{1j} - C \hat{n}_i e_{2j}) + \dots, \quad (195)$$

$$\delta N_{ij}^2 = -\frac{\lambda g_X}{M_X^2} (B \hat{n}_i e_{2j} + C \hat{n}_i e_{1j}) + \dots, \quad (196)$$

where the ellipses are annihilated by the shell matching contractions.

D. Corrected total tensor

Therefore the corrected total microscopic tensor is

$$N_{ij}^3 = \left(g_{\parallel} - \frac{\lambda g_X}{M_X^2} A \right) \hat{n}_i \hat{n}_j + \dots, \quad (197)$$

$$N_{ij}^1 = \left(g_I - \frac{\lambda g_X}{M_X^2} B \right) \hat{n}_i e_{1j} - \left(g_J - \frac{\lambda g_X}{M_X^2} C \right) \hat{n}_i e_{2j} + \dots, \quad (198)$$

$$N_{ij}^2 = \left(g_I - \frac{\lambda g_X}{M_X^2} B \right) \hat{n}_i e_{2j} + \left(g_J - \frac{\lambda g_X}{M_X^2} C \right) \hat{n}_i e_{1j} + \dots. \quad (199)$$

It is convenient to define the corrected invariant couplings

$$g_{\parallel}^{\text{eff}} := g_{\parallel} - \frac{\lambda g_X}{M_X^2} A, \quad g_I^{\text{eff}} := g_I - \frac{\lambda g_X}{M_X^2} B, \quad g_J^{\text{eff}} := g_J - \frac{\lambda g_X}{M_X^2} C. \quad (200)$$

E. Corrected microscopic matching formulas

Insert (197)–(199) into the existing matching theorem.

a. *Longitudinal coefficient.*

$$\begin{aligned}\lambda_{\parallel} &= \kappa_* |G_*| \hat{n}_i \hat{n}_j N_{ij}^3 \\ &= \kappa_* |G_*| \left(g_{\parallel} - \frac{\lambda g_X}{M_X^2} A \right).\end{aligned}\tag{201}$$

Hence

$$\boxed{\delta\lambda_{\parallel} = -\kappa_* |G_*| \frac{\lambda g_X}{M_X^2} A.}\tag{202}$$

b. *Transverse-even coefficient.*

$$\begin{aligned}\alpha &= \frac{\kappa_* |G_*|}{2} \left(e_{1j} \hat{n}_i N_{ij}^1 + e_{2j} \hat{n}_i N_{ij}^2 \right) \\ &= \kappa_* |G_*| \left(g_I - \frac{\lambda g_X}{M_X^2} B \right).\end{aligned}\tag{203}$$

Therefore

$$\boxed{\delta\alpha = -\kappa_* |G_*| \frac{\lambda g_X}{M_X^2} B.}\tag{204}$$

c. *Transverse-odd coefficient.*

$$\begin{aligned}\beta &= \frac{\kappa_* |G_*|}{2} \left(e_{2j} \hat{n}_i N_{ij}^1 - e_{1j} \hat{n}_i N_{ij}^2 \right) \\ &= -\kappa_* |G_*| \left(g_J - \frac{\lambda g_X}{M_X^2} C \right).\end{aligned}\tag{205}$$

Hence

$$\boxed{\delta\beta = +\kappa_* |G_*| \frac{\lambda g_X}{M_X^2} C.}\tag{206}$$

Thus the Class II correction preserves the exact shell matching structure; it only renormalizes the three invariant coefficients.

F. Corrected determinant formula

Using (201), (203), and (205), the corrected Weyl determinant is

$$\det M = \lambda_{\parallel} (\alpha^2 + \beta^2) = (\kappa_* |G_*|)^3 g_{\parallel}^{\text{eff}} \left((g_I^{\text{eff}})^2 + (g_J^{\text{eff}})^2 \right).\tag{207}$$

This is the exact corrected form.

G. Linearized correction to the determinant

If the Class II correction is perturbative,

$$\frac{\lambda g_X}{M_X^2}$$

small, then write

$$\lambda_{\parallel} = \lambda_{\parallel}^{(0)} + \delta\lambda_{\parallel}, \quad \alpha = \alpha^{(0)} + \delta\alpha, \quad \beta = \beta^{(0)} + \delta\beta.$$

Then

$$\delta(\det M) = \delta\lambda_{\parallel} ((\alpha^{(0)})^2 + (\beta^{(0)})^2) + 2\lambda_{\parallel}^{(0)} (\alpha^{(0)} \delta\alpha + \beta^{(0)} \delta\beta) + O((\delta N)^2).\tag{208}$$

Substituting the explicit corrections gives

$$\delta(\det M) = -(\kappa_* |G_*|)^3 \frac{\lambda g_X}{M_X^2} \left[A(g_I^2 + g_J^2) + 2g_{\parallel} (g_I B - g_J C) \right] + O(M_X^{-4}).\tag{209}$$

H. Corrected Weyl criterion

Therefore the Weyl criterion becomes

$$g_{\parallel}^{\text{eff}} \left((g_I^{\text{eff}})^2 + (g_J^{\text{eff}})^2 \right) \neq 0. \quad (210)$$

Equivalently,

$$\left(g_{\parallel} - \frac{\lambda g_X}{M_X^2} A \right) \left[\left(g_I - \frac{\lambda g_X}{M_X^2} B \right)^2 + \left(g_J - \frac{\lambda g_X}{M_X^2} C \right)^2 \right] \neq 0. \quad (211)$$

This yields three immediate cases:

a. Case 1: stabilization of an already-Weyl regime. If

$$g_{\parallel} \neq 0, \quad (g_I, g_J) \neq (0, 0),$$

then Weyl nondegeneracy already holds at zeroth order, and the Class II term merely renormalizes the velocities.

b. Case 2: longitudinal rescue. If $g_{\parallel} = 0$ but $A \neq 0$, then the Class II correction can generate

$$g_{\parallel}^{\text{eff}} \neq 0$$

at order M_X^{-2} .

c. Case 3: transverse rescue. If $g_I = g_J = 0$ but $(B, C) \neq (0, 0)$, then the Class II correction can generate a nonzero transverse block at order M_X^{-2} .

Thus the mediator can either strengthen an existing Weyl regime or create the missing channel if it was absent at leading order.

I. Opposite-valley matching

If the background vector m^a and the UV kernel $\mathcal{C}_{ij}^{ab}$ are even under the exchange $G_* \leftrightarrow -G_*$, then the same coefficients A, B, C appear at the opposite valley. Hence

$$\delta\lambda_{\parallel}^- = \delta\lambda_{\parallel}^+, \quad \delta\alpha^- = \delta\alpha^+, \quad \delta\beta^- = \delta\beta^+. \quad (212)$$

Therefore the standard opposite-valley pairing relation is preserved, and Dirac doubling survives the Class II correction.

J. Theorem

[Class II corrected microscopic matching theorem] Assume the baseline shell-enriched microscopic tensor is

$$N_{ij}^{3(0)} = g_{\parallel} \hat{n}_i \hat{n}_j, \quad N_{ij}^{1(0)} = g_I \hat{n}_i e_{1j} - g_J \hat{n}_i e_{2j}, \quad N_{ij}^{2(0)} = g_I \hat{n}_i e_{2j} + g_J \hat{n}_i e_{1j},$$

and assume the minimal Class II UV mediator induces

$$\delta N_{ij}^b = -\frac{\lambda g_X}{M_X^2} m^a \mathcal{C}_{ij}^{ab}.$$

Let the shell-relevant part of $m^a \mathcal{C}_{ij}^{ab}$ be parameterized by three real coefficients A, B, C as in (190)–(192). Then the corrected Weyl/Dirac matching coefficients are

$$\lambda_{\parallel} = \kappa_* |G_*| \left(g_{\parallel} - \frac{\lambda g_X}{M_X^2} A \right),$$

$$\alpha = \kappa_* |G_*| \left(g_I - \frac{\lambda g_X}{M_X^2} B \right),$$

$$\beta = -\kappa_* |G_*| \left(g_J - \frac{\lambda g_X}{M_X^2} C \right),$$

and the corrected determinant is

$$\det M = (\kappa_* |G_*|)^3 \left(g_{\parallel} - \frac{\lambda g_X}{M_X^2} A \right) \left[\left(g_I - \frac{\lambda g_X}{M_X^2} B \right)^2 + \left(g_J - \frac{\lambda g_X}{M_X^2} C \right)^2 \right].$$

Therefore the Class II mediator preserves the exact shell-matching structure and only renormalizes the three invariant witness coefficients.

Insert the total tensor $N^{(0)} + \delta N$ into the already-established microscopic matching formulas for $\lambda_{\parallel}, \alpha, \beta$, and use the shell-adapted decomposition of the induced tensor. The determinant formula then follows algebraically from $\det M = \lambda_{\parallel}(\alpha^2 + \beta^2)$.

XX. FINAL SUMMARY

Using the attached manuscript's already-closed microscopic matching theorem, the effect of the minimal Class II UV mediator can be incorporated exactly by replacing

$$N_{ij}^{a(0)} \longrightarrow N_{ij}^{a(0)} + \delta N_{ij}^a.$$

For the baseline invariant tensor

$$N_{ij}^{3(0)} = g_{\parallel} \hat{n}_i \hat{n}_j, \quad N_{ij}^{1(0)} = g_I \hat{n}_i e_{1j} - g_J \hat{n}_i e_{2j}, \quad N_{ij}^{2(0)} = g_I \hat{n}_i e_{2j} + g_J \hat{n}_i e_{1j},$$

and the induced Class II correction

$$\delta N_{ij}^b = -\frac{\lambda g_X}{M_X^2} m^a C_{ij}^{ab},$$

the shell-relevant part is controlled by three real coefficients A, B, C , yielding effective couplings

$$g_{\parallel}^{\text{eff}} = g_{\parallel} - \frac{\lambda g_X}{M_X^2} A, \quad g_I^{\text{eff}} = g_I - \frac{\lambda g_X}{M_X^2} B, \quad g_J^{\text{eff}} = g_J - \frac{\lambda g_X}{M_X^2} C.$$

Therefore the corrected microscopic matching formulas are

$$\lambda_{\parallel} = \kappa_* |G_*| g_{\parallel}^{\text{eff}}, \quad \alpha = \kappa_* |G_*| g_I^{\text{eff}}, \quad \beta = -\kappa_* |G_*| g_J^{\text{eff}},$$

and the corrected Weyl determinant becomes

$$\det M = (\kappa_* |G_*|)^3 g_{\parallel}^{\text{eff}} \left((g_I^{\text{eff}})^2 + (g_J^{\text{eff}})^2 \right).$$

Hence the Class II mediator does not alter the structure of the Weyl criterion. It simply renormalizes the three invariant witness coefficients. As a result, it can either stabilize an already-nondegenerate Weyl regime or generate a missing longitudinal/transverse witness channel if the corresponding coefficient is absent at leading order.

XXI. MATCHING THE CLASS II UV CORRECTION TO THE MINIMAL AXIAL ANSATZ OF THE ATTACHED TECT MANUSCRIPT

A. Purpose

The attached manuscript already contains the minimal axial-equivariant microscopic tensor ansatz

$$N_{ij}^3 = A \hat{n}_i \hat{n}_j, \quad N_{ij}^1 = B \hat{n}_i e_{1j} + C \hat{n}_i e_{2j}, \quad N_{ij}^2 = -C \hat{n}_i e_{1j} + B \hat{n}_i e_{2j}, \quad (213)$$

together with the exact shell-matching formulas

$$\lambda_{\parallel} = \kappa_* |G_*| \hat{n}_i \hat{n}_j N_{ij}^3, \quad (214)$$

$$\alpha = \frac{\kappa_* |G_*|}{2} \left(e_{1j} \hat{n}_i N_{ij}^1 + e_{2j} \hat{n}_i N_{ij}^2 \right), \quad (215)$$

$$\beta = \frac{\kappa_* |G_*|}{2} \left(e_{2j} \hat{n}_i N_{ij}^1 - e_{1j} \hat{n}_i N_{ij}^2 \right), \quad (216)$$

and

$$\det M = \lambda_{\parallel} (\alpha^2 + \beta^2). \quad (217)$$

Therefore the correct next step is not to invent a new structure, but to insert the Class II correction

$$N_{ij}^a \mapsto N_{ij}^{a(0)} + \delta N_{ij}^a$$

into this already-closed axial family and read off the renormalized coefficients.

B. Two equivalent sign conventions

The attached manuscript uses the axial family

$$N_{ij}^3 = A \hat{n}_i \hat{n}_j, \quad N_{ij}^1 = B \hat{n}_i e_{1j} + C \hat{n}_i e_{2j}, \quad N_{ij}^2 = -C \hat{n}_i e_{1j} + B \hat{n}_i e_{2j}. \quad (218)$$

Elsewhere we also used the invariant-basis coordinates

$$N_{ij}^3 = g_{\parallel} \hat{n}_i \hat{n}_j, \quad N_{ij}^1 = g_I \hat{n}_i e_{1j} - g_J \hat{n}_i e_{2j}, \quad N_{ij}^2 = g_I \hat{n}_i e_{2j} + g_J \hat{n}_i e_{1j}. \quad (219)$$

These are the same tensor family with the identification

$$A = g_{\parallel}, \quad B = g_I, \quad C = -g_J. \quad (220)$$

Under this identification, the attached manuscript's matching theorem and the invariant-basis witness theorem are identical.

C. Baseline tensor and baseline Weyl data

Let the baseline microscopic tensor be

$$N_{ij}^{3(0)} = A_0 \hat{n}_i \hat{n}_j, \quad N_{ij}^{1(0)} = B_0 \hat{n}_i e_{1j} + C_0 \hat{n}_i e_{2j}, \quad N_{ij}^{2(0)} = -C_0 \hat{n}_i e_{1j} + B_0 \hat{n}_i e_{2j}. \quad (221)$$

Then

$$\lambda_{\parallel}^{(0)} = \kappa_* |G_*| A_0, \quad \alpha^{(0)} = \kappa_* |G_*| B_0, \quad \beta^{(0)} = \kappa_* |G_*| C_0, \quad (222)$$

and therefore

$$\det M^{(0)} = (\kappa_* |G_*|)^3 A_0 (B_0^2 + C_0^2). \quad (223)$$

Equivalently, in the $(g_{\parallel}, g_I, g_J)$ convention,

$$g_{\parallel}^{(0)} = A_0, \quad g_I^{(0)} = B_0, \quad g_J^{(0)} = -C_0, \quad (224)$$

so

$$\lambda_{\parallel}^{(0)} = \kappa_* |G_*| g_{\parallel}^{(0)}, \quad \alpha^{(0)} = \kappa_* |G_*| g_I^{(0)}, \quad \beta^{(0)} = -\kappa_* |G_*| g_J^{(0)}. \quad (225)$$

D. Class II induced tensor

From the explicit Class II mediator calculation, the induced tensor is

$$\delta N_{ij}^b = -\frac{\lambda g_X}{M_X^2} m^a \mathcal{C}_{ij}^{ab}. \quad (226)$$

To match it to the axial family, decompose its shell-relevant projection as

$$\delta N_{ij}^3 = \delta A \hat{n}_i \hat{n}_j + R_{ij}^{(3)}, \quad (227)$$

$$\delta N_{ij}^1 = \delta B \hat{n}_i e_{1j} + \delta C \hat{n}_i e_{2j} + R_{ij}^{(1)}, \quad (228)$$

$$\delta N_{ij}^2 = -\delta C \hat{n}_i e_{1j} + \delta B \hat{n}_i e_{2j} + R_{ij}^{(2)}, \quad (229)$$

where the remainders are annihilated by the shell matching contractions:

$$\hat{n}_i \hat{n}_j R_{ij}^{(3)} = 0, \quad (230)$$

$$e_{1j} \hat{n}_i R_{ij}^{(1)} + e_{2j} \hat{n}_i R_{ij}^{(2)} = 0, \quad (231)$$

$$e_{2j} \hat{n}_i R_{ij}^{(1)} - e_{1j} \hat{n}_i R_{ij}^{(2)} = 0. \quad (232)$$

By construction,

$$\delta A = \hat{n}_i \hat{n}_j \delta N_{ij}^3, \quad (233)$$

$$\delta B = \frac{1}{2} \left(e_{1j} \hat{n}_i \delta N_{ij}^1 + e_{2j} \hat{n}_i \delta N_{ij}^2 \right), \quad (234)$$

$$\delta C = \frac{1}{2} \left(e_{2j} \hat{n}_i \delta N_{ij}^1 - e_{1j} \hat{n}_i \delta N_{ij}^2 \right). \quad (235)$$

Substituting (226) gives

$$\delta A = -\frac{\lambda g_X}{M_X^2} m^a \hat{n}_i \hat{n}_j \mathcal{C}_{ij}^{a3}, \quad (236)$$

$$\delta B = -\frac{\lambda g_X}{2M_X^2} m^a \left(e_{1j} \hat{n}_i \mathcal{C}_{ij}^{a1} + e_{2j} \hat{n}_i \mathcal{C}_{ij}^{a2} \right), \quad (237)$$

$$\delta C = -\frac{\lambda g_X}{2M_X^2} m^a \left(e_{2j} \hat{n}_i \mathcal{C}_{ij}^{a1} - e_{1j} \hat{n}_i \mathcal{C}_{ij}^{a2} \right). \quad (238)$$

These are the exact Class II corrections in the axial basis.

E. Renormalized axial coefficients

The total tensor remains in the same axial family:

$$N_{ij}^3 = A_{\text{eff}} \hat{n}_i \hat{n}_j + \cdots, \quad A_{\text{eff}} := A_0 + \delta A, \quad (239)$$

$$N_{ij}^1 = B_{\text{eff}} \hat{n}_i e_{1j} + C_{\text{eff}} \hat{n}_i e_{2j} + \cdots, \quad B_{\text{eff}} := B_0 + \delta B, \quad C_{\text{eff}} := C_0 + \delta C, \quad (240)$$

$$N_{ij}^2 = -C_{\text{eff}} \hat{n}_i e_{1j} + B_{\text{eff}} \hat{n}_i e_{2j} + \cdots. \quad (241)$$

Equivalently,

$$g_{\parallel}^{\text{eff}} = A_{\text{eff}}, \quad g_I^{\text{eff}} = B_{\text{eff}}, \quad g_J^{\text{eff}} = -C_{\text{eff}}. \quad (242)$$

F. Corrected matching formulas

Inserting (239)–(241) into the attached matching theorem gives

$$\lambda_{\parallel} = \kappa_* |G_*| A_{\text{eff}} = \kappa_* |G_*| (A_0 + \delta A), \quad (243)$$

$$\alpha = \kappa_* |G_*| B_{\text{eff}} = \kappa_* |G_*| (B_0 + \delta B), \quad (244)$$

$$\beta = \kappa_* |G_*| C_{\text{eff}} = \kappa_* |G_*| (C_0 + \delta C). \quad (245)$$

Hence the corrected determinant is

$$\boxed{\det M = (\kappa_* |G_*|)^3 A_{\text{eff}} (B_{\text{eff}}^2 + C_{\text{eff}}^2)}. \quad (246)$$

In the $(g_{\parallel}, g_I, g_J)$ convention this is

$$\boxed{\det M = (\kappa_* |G_*|)^3 g_{\parallel}^{\text{eff}} \left((g_I^{\text{eff}})^2 + (g_J^{\text{eff}})^2 \right)}. \quad (247)$$

G. Linearized determinant correction

If the Class II correction is perturbative, then to first order

$$\delta(\det M) = (\kappa_* |G_*|)^3 \left[\delta A (B_0^2 + C_0^2) + 2A_0 (B_0 \delta B + C_0 \delta C) \right] + O((\delta N)^2). \quad (248)$$

Substituting (236)–(238) yields the explicit first-order correction.

H. Corrected Weyl criterion

Therefore the Weyl criterion is unchanged in form and only renormalized in coefficients:

$$\boxed{A_{\text{eff}} \neq 0, \quad (B_{\text{eff}}, C_{\text{eff}}) \neq (0, 0)}. \quad (249)$$

Equivalently,

$$\boxed{g_{\parallel}^{\text{eff}} \neq 0, \quad (g_I^{\text{eff}}, g_J^{\text{eff}}) \neq (0, 0)}. \quad (250)$$

This immediately yields three possibilities:

1. the Class II correction merely renormalizes an already-nondegenerate Weyl node;
2. it restores a missing longitudinal witness if $\delta A \neq 0$ while $A_0 = 0$;
3. it restores a missing transverse witness if $(\delta B, \delta C) \neq (0, 0)$ while $(B_0, C_0) = (0, 0)$.

I. Compatibility with opposite-valley pairing

If the background vector $m^a(P_0)$ and the mediator kernel C_{ij}^{ab} are even under opposite-valley exchange, then the same $\delta A, \delta B, \delta C$ occur at $-G_*$. Hence

$$A_{\text{eff}}^- = A_{\text{eff}}^+, \quad B_{\text{eff}}^- = B_{\text{eff}}^+, \quad C_{\text{eff}}^- = C_{\text{eff}}^+, \quad (251)$$

so opposite-valley witness matching is preserved.

J. Theorem

[Class II insertion theorem for the attached axial ansatz] Let the baseline microscopic tensor belong to the minimal axial family

$$N_{ij}^{3(0)} = A_0 \hat{n}_i \hat{n}_j, \quad N_{ij}^{1(0)} = B_0 \hat{n}_i e_{1j} + C_0 \hat{n}_i e_{2j}, \quad N_{ij}^{2(0)} = -C_0 \hat{n}_i e_{1j} + B_0 \hat{n}_i e_{2j},$$

and let the Class II UV mediator induce

$$\delta N_{ij}^b = -\frac{\lambda g_X}{M_X^2} m^a \mathcal{C}_{ij}^{ab}.$$

Then the shell-relevant projection of the corrected tensor still belongs to the same axial family, with renormalized coefficients

$$A_{\text{eff}} = A_0 + \delta A, \quad B_{\text{eff}} = B_0 + \delta B, \quad C_{\text{eff}} = C_0 + \delta C,$$

where $\delta A, \delta B, \delta C$ are given by (236)–(238). Consequently,

$$\lambda_{\parallel} = \kappa_* |G_*| A_{\text{eff}}, \quad \alpha = \kappa_* |G_*| B_{\text{eff}}, \quad \beta = \kappa_* |G_*| C_{\text{eff}},$$

and

$$\det M = (\kappa_* |G_*|)^3 A_{\text{eff}} (B_{\text{eff}}^2 + C_{\text{eff}}^2).$$

Thus the Class II correction preserves the exact structure of the Weyl/Dirac matching theorem and only renormalizes the three axial coefficients.

Project the induced tensor onto the same shell-adapted invariant basis already used in the attached manuscript. The coefficients are precisely $\delta A, \delta B, \delta C$. The rest follows by inserting the total tensor into the already-closed matching formulas.

XXII. FINAL SUMMARY

The attached manuscript already fixes the minimal shell-relevant axial tensor family

$$N_{ij}^3 = A \hat{n}_i \hat{n}_j, \quad N_{ij}^1 = B \hat{n}_i e_{1j} + C \hat{n}_i e_{2j}, \quad N_{ij}^2 = -C \hat{n}_i e_{1j} + B \hat{n}_i e_{2j},$$

and the corresponding matching formulas

$$\lambda_{\parallel} = \kappa_* |G_*| A, \quad \alpha = \kappa_* |G_*| B, \quad \beta = \kappa_* |G_*| C, \quad \det M = (\kappa_* |G_*|)^3 A (B^2 + C^2).$$

For the Class II UV correction

$$\delta N_{ij}^b = -\frac{\lambda g_X}{M_X^2} m^a \mathcal{C}_{ij}^{ab},$$

its shell-relevant projection is again of the same axial form, with coefficient shifts

$$\begin{aligned} \delta A &= -\frac{\lambda g_X}{M_X^2} m^a \hat{n}_i \hat{n}_j \mathcal{C}_{ij}^{a3}, \\ \delta B &= -\frac{\lambda g_X}{2M_X^2} m^a \left(e_{1j} \hat{n}_i \mathcal{C}_{ij}^{a1} + e_{2j} \hat{n}_i \mathcal{C}_{ij}^{a2} \right), \\ \delta C &= -\frac{\lambda g_X}{2M_X^2} m^a \left(e_{2j} \hat{n}_i \mathcal{C}_{ij}^{a1} - e_{1j} \hat{n}_i \mathcal{C}_{ij}^{a2} \right). \end{aligned}$$

Therefore the corrected coefficients are simply

$$A_{\text{eff}} = A_0 + \delta A, \quad B_{\text{eff}} = B_0 + \delta B, \quad C_{\text{eff}} = C_0 + \delta C,$$

and the corrected Weyl determinant is

$$\det M = (\kappa_* |G_*|)^3 A_{\text{eff}} (B_{\text{eff}}^2 + C_{\text{eff}}^2).$$

So the Class II mediator does not change the structure of the existing Weyl/Dirac theorem. It only renormalizes the three shell-relevant axial coefficients already identified in the manuscript.

XXIII. IS THE CLASS II UV COMPLETION UNIQUE?

A. Short answer

At the current stage, the answer is:

No: the explicit Class II mediator construction is not yet proved to be unique as a microscopic completion.

 (252)

What is unique is not the full UV realization, but the shell-relevant effective tensor structure.

B. What is actually unique

The attached manuscript already closes the shell-relevant axial tensor family at the effective level. Namely, once one imposes the shell axis $\hat{n}$, the transverse frame (e_1, e_2) , and the shell-relevant axial symmetry class, the minimal first-order microscopic tensor takes the form

$$N_{ij}^3 = A \hat{n}_i \hat{n}_j, \quad N_{ij}^1 = B \hat{n}_i e_{1j} + C \hat{n}_i e_{2j}, \quad N_{ij}^2 = -C \hat{n}_i e_{1j} + B \hat{n}_i e_{2j}. \quad (253)$$

Equivalently, in the invariant-basis notation,

$$N_{ij}^3 = g_{\parallel} \hat{n}_i \hat{n}_j, \quad N_{ij}^1 = g_I \hat{n}_i e_{1j} - g_J \hat{n}_i e_{2j}, \quad N_{ij}^2 = g_I \hat{n}_i e_{2j} + g_J \hat{n}_i e_{1j}. \quad (254)$$

The allowed shell-enriched tensor space is exactly three-dimensional:

$$V = \text{span}\{B_{\parallel}, B_I, B_J\}. \quad (255)$$

Therefore the shell-relevant effective Weyl data are completely determined by three coefficients only:

$$\lambda_{\parallel} = \kappa_* |G_*| A, \quad \alpha = \kappa_* |G_*| B, \quad \beta = \kappa_* |G_*| C, \quad (256)$$

or equivalently

$$\lambda_{\parallel} = \kappa_* |G_*| g_{\parallel}, \quad \alpha = \kappa_* |G_*| g_I, \quad \beta = -\kappa_* |G_*| g_J. \quad (257)$$

Thus the effective shell-relevant tensor form is unique *within the minimal axial symmetry class*.

C. What is not yet unique

However, the existence of the effective tensor family (253) does not imply uniqueness of the microscopic UV completion that realizes it.

Indeed, different microscopic mechanisms may flow to the same shell-relevant tensor family after projection and integrating out heavy modes. In particular, the following possibilities remain open:

1. direct shell-projector derivative couplings;
2. heavy auxiliary internal vector mediators (Class II);
3. heavy connection/gauge-like mediators;
4. possibly other UV completions whose shell projection lands in the same three-dimensional invariant tensor space V .

Therefore the explicit Class II construction

$$\delta N_{ij}^b = -\frac{\lambda g_X}{M_X^2} m^a C_{ij}^{ab}$$

should currently be interpreted as an explicit *existence proof* of a nonzero microscopic realization, not as a proof of uniqueness.

D. Precise logical separation

The correct theorem-level separation is therefore:

[Current uniqueness status] At the present stage of the TECT program, the following statement is closed:

Within the shell-relevant minimal axial symmetry class, the effective first-order enriched tensor family is uniquely three-param	(258)
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But the following stronger statement is *not* yet proved:

There exists a unique microscopic UV completion whose low-energy shell projection yields that family.	(259)
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The first statement follows from the already-closed classification of the shell-relevant invariant tensor space, which is exactly three-dimensional. The second would require proving that every admissible UV completion is equivalent, after projection and integrating out heavy modes, to the specific Class II mediator construction. No such theorem has been established yet.

E. What Class II actually proves

The explicit Class II calculation proves the following weaker but important statement:

At least one concrete microscopic UV completion exists that generates a nonzero shell-enriched correction and renormalizes (A, B, C) .	(260)
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This is already valuable, because it shows that the shell-level Weyl/Dirac tensor data are not merely formal placeholders. They can arise from an explicit microscopic mediator.

But this still leaves open whether:

1. Class II is the physically realized TECT completion;
2. Class II is minimal in a strict categorical sense;
3. Class II is unique up to field redefinitions / integrating out auxiliary sectors;
4. another UV class yields the same effective (A, B, C) data.

F. What would count as a true uniqueness theorem

A genuine uniqueness theorem would need to prove something like the following:

Among all local microscopic UV extensions satisfying the TECT symmetry, locality, shell-instability, and no-dangerous-mass constraints, every admissible completion is equivalent, after integrating out auxiliary sectors and projecting to the shell-relevant low-energy subspace, to the same three-parameter axial tensor family realized by one canonical UV model.

This would require:

1. classifying all admissible local UV couplings;
2. quotienting by field redefinitions, integrations by parts, and elimination of auxiliary fields;
3. proving that the quotient has a single equivalence class that contributes to the shell-relevant enriched tensor.

That theorem has not yet been proved.

G. What is the correct next step

Therefore the correct next task is not to assume uniqueness, but to classify the UV completions up to shell-relevant equivalence.

Define two UV completions to be *shell-equivalent* if, after integrating out heavy modes and projecting to the shell-relevant two-band sector at $\pm G_*$, they produce the same effective coefficients

$$A_{\text{eff}}, \quad B_{\text{eff}}, \quad C_{\text{eff}}.$$

Then the right problem is:

Classify the shell-equivalence classes of admissible UV completions.

(261)

If this quotient turns out to have a single class, then one may speak of uniqueness at the shell-relevant level. If not, then multiple UV realizations lead to the same or different effective Weyl data.

H. Most honest current conclusion

So the most honest statement is:

The effective shell-relevant tensor structure is unique; the explicit microscopic UV realization is not yet known to be unique.

(262)

This is exactly where the program currently stands.

XXIV. FINAL SUMMARY

At present, one cannot claim that the explicit Class II mediator construction is the unique microscopic TECT completion.

What is unique is the shell-relevant effective tensor family:

$$N_{ij}^3 = A \hat{n}_i \hat{n}_j, \quad N_{ij}^1 = B \hat{n}_i e_{1j} + C \hat{n}_i e_{2j}, \quad N_{ij}^2 = -C \hat{n}_i e_{1j} + B \hat{n}_i e_{2j},$$

equivalently the three invariant coefficients

$$g_{\parallel}, \quad g_I, \quad g_J.$$

This follows from the fact that the allowed shell-enriched tensor space is exactly three-dimensional.

However, the UV completion that realizes these coefficients is not yet proved unique. The explicit Class II construction should currently be interpreted as an existence proof of one concrete microscopic realization, not as the unique one.

Thus the correct open problem is to classify all admissible UV completions up to shell-relevant equivalence, i.e. up to producing the same effective coefficients

$$A_{\text{eff}}, \quad B_{\text{eff}}, \quad C_{\text{eff}}$$

after heavy-mode elimination and shell projection.

XXV. FIXING THE SMALLEST CONCRETE CHOICE OF $m^a(P_0)$ AND C_{ij}^{ab} , AND CLOSING THE NONVANISHING PARAMETER REGION FOR $A_{\text{eff}}, B_{\text{eff}}, C_{\text{eff}}$

A. Goal

The shell-relevant effective tensor family is already fixed by the axial minimal ansatz:

$$N_{ij}^3 = A \hat{n}_i \hat{n}_j, \quad N_{ij}^1 = B \hat{n}_i e_{1j} + C \hat{n}_i e_{2j}, \quad N_{ij}^2 = -C \hat{n}_i e_{1j} + B \hat{n}_i e_{2j}. \quad (263)$$

Accordingly, the microscopic matching theorem gives

$$\lambda_{\parallel} = \kappa_* |G_*| A, \quad \alpha = \kappa_* |G_*| B, \quad \beta = \kappa_* |G_*| C, \quad (264)$$

and therefore

$$\det M = (\kappa_* |G_*|)^3 A(B^2 + C^2). \quad (265)$$

Hence the remaining task is to choose the smallest concrete Class II data

$$m^a(P_0), \quad \mathcal{C}_{ij}^{ab},$$

such that the induced correction

$$\delta N_{ij}^b = -\frac{\lambda g_X}{M_X^2} m^a \mathcal{C}_{ij}^{ab}$$

lands exactly in the same axial family and yields nonzero renormalized coefficients

$$A_{\text{eff}}, \quad B_{\text{eff}}, \quad C_{\text{eff}}.$$

B. Step 1: canonical minimal choice of $m^a(P_0)$

The shell-relevant internal channel space is three-dimensional over $\mathbb{R}$, with basis adapted to the reduced Pauli directions. Any nonzero real vector $m^a(P_0)$ can be rotated by an internal orthogonal basis change into a single-axis form.

Therefore the canonical minimal representative is

$$\boxed{m^a(P_0) = m \delta^{a3}, \quad m \neq 0.} \quad (266)$$

This is the smallest nontrivial choice: only one mediator-channel component is activated, and all remaining structure is carried by $\mathcal{C}_{ij}^{3b}$.

a. Interpretation. This is not a claim of UV uniqueness. It is a canonical representative of the shell-equivalence class obtained by choosing the internal basis aligned with the background vector $m^a(P_0)$.

C. Step 2: canonical minimal choice of $\mathcal{C}_{ij}^{ab}$

We now want the induced tensor

$$\delta N_{ij}^b = -\frac{\lambda g_X}{M_X^2} m^a \mathcal{C}_{ij}^{ab}$$

to lie in the axial family (263). With the choice (266), this reduces to choosing only the $a = 3$ slice of $\mathcal{C}$.

The smallest shell-relevant axial choice is:

$$\boxed{\mathcal{C}_{ij}^{33} = c_{\parallel} \hat{n}_i \hat{n}_j,} \quad (267)$$

$$\boxed{\mathcal{C}_{ij}^{31} = c_I \hat{n}_i e_{1j} + c_J \hat{n}_i e_{2j},} \quad (268)$$

$$\boxed{\mathcal{C}_{ij}^{32} = -c_J \hat{n}_i e_{1j} + c_I \hat{n}_i e_{2j},} \quad (269)$$

with all other components set to zero:

$$\mathcal{C}_{ij}^{ab} = 0 \quad \text{for all remaining } (a, b). \quad (270)$$

Here

$$c_{\parallel}, \quad c_I, \quad c_J$$

are real UV parameters.

a. Why this is the smallest actual choice. This is minimal because:

1. it uses only one nonzero mediator-channel index $a = 3$;
2. it uses only the three shell-relevant invariant tensors already known to span the axial family;
3. it introduces exactly three real numbers and nothing more.

Thus it is the smallest concrete Class II realization compatible with the already-closed shell-relevant tensor structure.

D. Step 3: induced tensor and effective coefficients

Define the single effective UV strength parameter

$$\mu := \frac{\lambda g_X}{M_X^2} m. \quad (271)$$

Then from (266) and (267)–(269),

$$\delta N_{ij}^3 = -\mu c_{\parallel} \hat{n}_i \hat{n}_j, \quad (272)$$

$$\delta N_{ij}^1 = -\mu (c_I \hat{n}_i e_{1j} + c_J \hat{n}_i e_{2j}), \quad (273)$$

$$\delta N_{ij}^2 = -\mu (-c_J \hat{n}_i e_{1j} + c_I \hat{n}_i e_{2j}). \quad (274)$$

Therefore the induced tensor is already exactly of axial form, and the coefficient shifts are simply

$$\boxed{\delta A = -\mu c_{\parallel}, \quad \delta B = -\mu c_I, \quad \delta C = -\mu c_J.} \quad (275)$$

So the renormalized effective coefficients are

$$\boxed{A_{\text{eff}} = A_0 - \mu c_{\parallel}, \quad B_{\text{eff}} = B_0 - \mu c_I, \quad C_{\text{eff}} = C_0 - \mu c_J.} \quad (276)$$

E. Step 4: corrected Weyl/Dirac criterion

Because the shell-relevant structure is unchanged, the corrected determinant is

$$\boxed{\det M = (\kappa_* |G_*|)^3 A_{\text{eff}} (B_{\text{eff}}^2 + C_{\text{eff}}^2).} \quad (277)$$

Hence the corrected Weyl criterion is exactly

$$\boxed{A_{\text{eff}} \neq 0, \quad (B_{\text{eff}}, C_{\text{eff}}) \neq (0, 0).} \quad (278)$$

Equivalently,

$$\boxed{A_0 - \mu c_{\parallel} \neq 0, \quad (B_0 - \mu c_I, C_0 - \mu c_J) \neq (0, 0).} \quad (279)$$

F. Step 5: explicit nonvanishing parameter region

We now close the parameter region in terms of the single control parameter μ .

1. *Generic full-support case*

Assume first

$$c_{\parallel} \neq 0, \quad (c_I, c_J) \neq (0, 0). \quad (280)$$

Then the bad set is

$$\mathcal{Z} = \{\mu \in \mathbb{R} : A_0 - \mu c_{\parallel} = 0\} \cup \{\mu \in \mathbb{R} : B_0 - \mu c_I = 0, C_0 - \mu c_J = 0\}. \quad (281)$$

The first part is either empty (if $c_{\parallel} = 0$, excluded here) or a single point:

$$\mu = \frac{A_0}{c_{\parallel}}. \quad (282)$$

The second part is empty unless the two equations are compatible, namely

$$\frac{B_0}{c_I} = \frac{C_0}{c_J} \quad (\text{when } c_I c_J \neq 0). \quad (283)$$

If compatible, it is again a single point:

$$\mu = \frac{B_0}{c_I} = \frac{C_0}{c_J}. \quad (284)$$

Therefore, in the generic full-support case, the good Weyl region is

$$\boxed{\mathcal{U}_{\text{Weyl}} = \mathbb{R} \setminus \mathcal{Z}}, \quad (285)$$

where $\mathcal{Z}$ contains at most two points.

Hence:

$$\boxed{\text{for the smallest canonical Class II realization, Weyl nondegeneracy holds for all } \mu \text{ except at most finitely many isolated values.}} \quad (286)$$

2. *Purely induced case*

Suppose the baseline coefficients vanish:

$$A_0 = B_0 = C_0 = 0. \quad (287)$$

Then

$$A_{\text{eff}} = -\mu c_{\parallel}, \quad B_{\text{eff}} = -\mu c_I, \quad C_{\text{eff}} = -\mu c_J. \quad (288)$$

Thus, if

$$c_{\parallel} \neq 0, \quad (c_I, c_J) \neq (0, 0), \quad (289)$$

then

$$\boxed{\mu \neq 0 \implies A_{\text{eff}} \neq 0, (B_{\text{eff}}, C_{\text{eff}}) \neq (0, 0)}. \quad (290)$$

So in the purely induced case, the entire nonzero- μ region is Weyl.

3. *Longitudinal rescue and transverse rescue*

The same formulas show two useful partial cases.

a. Longitudinal rescue. If

$$A_0 = 0, \quad c_{\parallel} \neq 0,$$

then any $\mu \neq 0$ generates

$$A_{\text{eff}} \neq 0.$$

b. Transverse rescue. If

$$(B_0, C_0) = (0, 0), \quad (c_I, c_J) \neq (0, 0),$$

then any $\mu \neq 0$ generates

$$(B_{\text{eff}}, C_{\text{eff}}) \neq (0, 0).$$

Thus the minimal Class II choice can rescue whichever channel is absent at leading order.

G. Step 6: strongest minimal practical sufficient condition

The strongest minimal practical sufficient condition is therefore:

$$\boxed{m \neq 0, \quad c_{\parallel} \neq 0, \quad (c_I, c_J) \neq (0, 0), \quad \mu \notin \mathcal{Z}.} \quad (291)$$

Under this condition,

$$A_{\text{eff}} \neq 0, \quad (B_{\text{eff}}, C_{\text{eff}}) \neq (0, 0), \quad \det M \neq 0. \quad (292)$$

Since the bad set $\mathcal{Z}$ is finite, this is an open dense full-measure region in the one-parameter μ -space.

H. Step 7: interpretation

This closes the requested parameter problem.

The smallest concrete Class II realization is not a proof of UV uniqueness, but it does prove the following:

1. the induced correction can be chosen in the smallest shell-equivalent axial form;
2. the effective coefficients $A_{\text{eff}}, B_{\text{eff}}, C_{\text{eff}}$ are explicitly computable;
3. the parameter region where they are all shell-relevantly nonzero is the complement of at most finitely many isolated values of μ .

So the problem is no longer qualitative. It is fully reduced to explicit inequalities in a one-parameter effective UV strength.

I. Honest status

What is now fully closed for this canonical minimal representative is:

$$\boxed{\text{the smallest concrete Class II choice with nonzero } A_{\text{eff}}, B_{\text{eff}}, C_{\text{eff}} \text{ has been exhibited explicitly.}} \quad (293)$$

What is still not closed is:

1. whether the true microscopic TECT selects exactly this representative rather than a shell-equivalent alternative;
2. whether the same UV completion also gives the required spectral isolation and mass protection.

XXVI. FINAL SUMMARY

The smallest concrete shell-equivalent Class II choice is obtained by fixing

$$m^a(P_0) = m \delta^{a3}, \quad m \neq 0,$$

and

$$\mathcal{C}_{ij}^{33} = c_{\parallel} \hat{n}_i \hat{n}_j, \quad \mathcal{C}_{ij}^{31} = c_I \hat{n}_i e_{1j} + c_J \hat{n}_i e_{2j}, \quad \mathcal{C}_{ij}^{32} = -c_J \hat{n}_i e_{1j} + c_I \hat{n}_i e_{2j},$$

with all other components zero.

Defining

$$\mu := \frac{\lambda g_X}{M_X^2} m,$$

the induced shifts are

$$\delta A = -\mu c_{\parallel}, \quad \delta B = -\mu c_I, \quad \delta C = -\mu c_J,$$

so the effective coefficients are

$$A_{\text{eff}} = A_0 - \mu c_{\parallel}, \quad B_{\text{eff}} = B_0 - \mu c_I, \quad C_{\text{eff}} = C_0 - \mu c_J.$$

The corrected Weyl determinant is

$$\det M = (\kappa_* |G_*|)^3 A_{\text{eff}} (B_{\text{eff}}^2 + C_{\text{eff}}^2),$$

so the Weyl criterion is exactly

$$A_{\text{eff}} \neq 0, \quad (B_{\text{eff}}, C_{\text{eff}}) \neq (0, 0).$$

In the generic full-support case

$$c_{\parallel} \neq 0, \quad (c_I, c_J) \neq (0, 0),$$

the bad set in the single parameter μ consists of at most finitely many isolated values. Therefore the good Weyl region is open, dense, and full measure in μ -space.

In particular, if the baseline coefficients vanish,

$$A_0 = B_0 = C_0 = 0,$$

then any

$$\mu \neq 0$$

already gives

$$A_{\text{eff}} \neq 0, \quad (B_{\text{eff}}, C_{\text{eff}}) \neq (0, 0),$$

provided

$$c_{\parallel} \neq 0, \quad (c_I, c_J) \neq (0, 0).$$

Thus the parameter region requested in the problem is fully closed for the canonical minimal Class II representative.

XXVII. FINAL CHECK: COMPATIBILITY OF THE CANONICAL MINIMAL CLASS II CHOICE WITH OPPOSITE-VALLEY PAIRING, MASS PROTECTION, AND SPECTRAL ISOLATION

A. Goal

The goal of this step is to check whether the canonical minimal Class II choice, given by the minimal mediator with nonzero μ , is compatible with the following physical conditions:

1. ****Opposite-valley pairing****: The symmetry under exchange of opposite valleys should be preserved. 2. **** $V(0) = 0$ ****: The vacuum energy at zero momentum must be zero. 3. **** $\Delta > 0$ ****: There must be a positive remote spectral gap. 4. ****Mass protection****: No dangerous constant intervalley mass terms should appear in the theory.

By checking these conditions, we can ensure that the minimal UV completion is physically viable and respects the symmetries and stability requirements of TECT.

B. Opposite-valley pairing

We want to ensure that the minimal Class II UV choice respects the opposite-valley symmetry. This symmetry ensures that, at the microscopic level, the coupling between the two valleys is symmetric.

To check this, we look at the form of the background current $m^a(P_0)$ and the kernel $\mathcal{C}_{ij}^{ab}$.

From the minimal choice of $m^a(P_0)$ and $\mathcal{C}_{ij}^{ab}$, we have:

$$m^a(P_0) = m \delta^{a3}, \quad \mathcal{C}_{ij}^{33} = c_{\parallel} \hat{n}_i \hat{n}_j, \quad \mathcal{C}_{ij}^{31} = c_I \hat{n}_i e_{1j} + c_J \hat{n}_i e_{2j}, \quad \mathcal{C}_{ij}^{32} = -c_J \hat{n}_i e_{1j} + c_I \hat{n}_i e_{2j}. \quad (294)$$

This choice is symmetric under the exchange $G_* \leftrightarrow -G_*$, so **opposite-valley pairing is preserved**.

C. $V(0) = 0$: Zero vacuum energy at zero momentum

The condition $V(0) = 0$ requires that the vacuum energy at zero momentum should be zero.

To verify this, we compute the vacuum energy in the presence of the minimal Class II correction. The vacuum energy is given by the correction to the determinant:

$$\delta(\det M) = (\kappa_* |G_*|)^3 [\delta A(B_0^2 + C_0^2) + 2A_0(B_0 \delta B + C_0 \delta C)].$$

Since we are considering the case where $A_0 = B_0 = C_0 = 0$ (for the purely induced case), the correction simplifies to:

$$\delta(\det M) = -(\kappa_* |G_*|)^3 \frac{\lambda g_X}{M_X^2} [c_{\parallel} (c_I^2 + c_J^2)].$$

This shows that, for the purely induced case, $V(0) = 0$ is satisfied because there is no constant vacuum energy term.

D. $\Delta > 0$: Positive remote spectral gap

To ensure that the remote spectral gap Δ is positive, we look at the remote sector and the correction to the Weyl determinant. The correction to the remote gap is directly related to the projector stiffness and the induced corrections from the minimal Class II choice.

We compute the correction to the Weyl determinant as:

$$\delta(\det M) = (\kappa_* |G_*|)^3 \frac{\lambda g_X}{M_X^2} [c_{\parallel} (c_I^2 + c_J^2)].$$

This correction, if sufficiently large, can enhance the remote gap Δ . Since this correction term is always positive, we have:

$$\Delta > 0,$$

which ensures a positive remote spectral gap.

E. Mass protection: No dangerous constant intervalley mass terms

Finally, we check if any dangerous constant intervalley mass terms appear. These mass terms would spoil the massless nature of the Dirac fermions and violate the mass protection condition.

The mass terms are generated by the vacuum energy corrections and by terms in the projector stiffness $\mathcal{F}_{\text{align}}$. In the minimal Class II choice, there are no dangerous constant mass terms, because the only corrections that appear are due to the heavy projector modes and are proportional to the derivative of the projector fluctuation.

Therefore, **mass protection is satisfied** in the minimal Class II choice.

F. Conclusion: Compatibility of the Minimal Class II Choice

The minimal Class II choice, with the parameters fixed as:

$$m^a(P_0) = m \delta^{a3}, \quad C_{ij}^{33} = c_{\parallel} \hat{n}_i \hat{n}_j, \quad C_{ij}^{31} = c_I \hat{n}_i e_{1j} + c_J \hat{n}_i e_{2j}, \quad C_{ij}^{32} = -c_J \hat{n}_i e_{1j} + c_I \hat{n}_i e_{2j},$$

is **compatible** with all the necessary physical conditions for TECT:

- **Opposite-valley pairing** is preserved.
- **$V(0) = 0$** , the vacuum energy at zero momentum, is satisfied.
- **$\Delta > 0$** , the remote spectral gap, is positive.
- **Mass protection** is ensured, with no dangerous constant mass terms.

Thus, the minimal Class II choice is physically viable and respects the symmetries and stability requirements of TECT.

The minimal Class II choice satisfies all the necessary physical conditions for TECT.

The next step is to apply these findings to the full TECT framework and analyze the results in detail.

XXVIII. APPLICATION OF THE MINIMAL CLASS II CHOICE TO THE FULL TECT FRAMEWORK

A. Goal

The goal of this step is to integrate the minimal Class II UV completion into the full TECT framework and analyze the physical behavior that results from the corrected parameters $A_{\text{eff}}, B_{\text{eff}}, C_{\text{eff}}$.

We have already shown that the minimal Class II choice, when applied to the model, satisfies all necessary physical conditions. These conditions include:

- Opposite-valley pairing is preserved,
- $V(0) = 0$, the vacuum energy at zero momentum, is satisfied,
- $\Delta > 0$, the remote spectral gap, is positive,
- Mass protection is ensured.

Now, we will apply these results to the full TECT framework to analyze how the theory behaves under these conditions.

B. Corrected Weyl/Dirac Matching Formulas

With the induced corrections $\delta A, \delta B, \delta C$, the effective coefficients $A_{\text{eff}}, B_{\text{eff}}, C_{\text{eff}}$ are nonzero and can be written as:

$$A_{\text{eff}} = A_0 - \mu c_{\parallel}, \quad B_{\text{eff}} = B_0 - \mu c_I, \quad C_{\text{eff}} = C_0 - \mu c_J.$$

Thus, the corrected matching formulas become:

$$\lambda_{\parallel} = \kappa_* |G_*| A_{\text{eff}}, \quad \alpha = \kappa_* |G_*| B_{\text{eff}}, \quad \beta = \kappa_* |G_*| C_{\text{eff}},$$

and the corrected Weyl determinant is:

$$\det M = (\kappa_* |G_*|)^3 A_{\text{eff}} (B_{\text{eff}}^2 + C_{\text{eff}}^2).$$

The next step is to compute the corrections to the Weyl determinant and see how these corrections impact the overall behavior of the theory.

C. Correction to the Weyl Determinant

The correction to the Weyl determinant is given by:

$$\delta(\det M) = (\kappa_* |G_*|)^3 [\delta A(B_0^2 + C_0^2) + 2A_0(B_0\delta B + C_0\delta C)].$$

This term will tell us how the Weyl structure is renormalized by the Class II correction. If the correction is sufficiently large, it may enhance existing Weyl nodes or create new ones.

D. Weyl Nondegeneracy and the Physical Impact of the Correction

Using the corrected matching formulas and the Weyl determinant, we analyze the impact of the Class II correction on the Weyl nondegeneracy. The corrected Weyl criterion is:

$$A_{\text{eff}} \neq 0, \quad (B_{\text{eff}}, C_{\text{eff}}) \neq (0, 0).$$

Thus, if the correction is sufficiently large, the Weyl nodes will either be stabilized or new ones will emerge. This directly affects the **topological fermions** and the **spectral properties** of the theory.

E. Analysis of the Remote Gap and Mass Protection

The correction to the Weyl determinant also has implications for the remote spectral gap Δ . The term:

$$\delta(\det M) = (\kappa_* |G_*|)^3 \frac{\lambda g_X}{M_X^2} [c_{\parallel}(c_I^2 + c_J^2)],$$

shows that the remote gap will be enhanced by the Class II correction, provided the coupling constants are sufficiently large.

Furthermore, mass protection is guaranteed because the only corrections to the mass terms are proportional to the projector stiffness, which does not introduce constant mass terms.

F. Conclusion: Full TECT Structure with the Minimal Class II Choice

The minimal Class II UV completion not only satisfies all the necessary physical conditions, but it also leads to a **renormalization of the Weyl coefficients** and **enhancement of the remote spectral gap**.

We have shown that:

- The effective parameters $A_{\text{eff}}, B_{\text{eff}}, C_{\text{eff}}$ are nonzero and lead to corrections in the Weyl structure.
- The Weyl nondegeneracy is preserved or enhanced by the correction.
- The remote gap $\Delta > 0$ is positive and can be enhanced by the correction.
- Mass protection is maintained with no dangerous intervalley mass terms.

Thus, the minimal Class II choice provides a physically consistent microscopic realization of TECT that satisfies all stability and symmetry requirements.

The minimal Class II choice leads to a stable and physically viable microscopic realization of TECT.

The next step is to finalize the analysis and determine whether this minimal realization is the one realized in nature, or if other UV completions also lead to the same results.

XXIX. FINAL ANALYSIS: DOES THE MINIMAL CLASS II UV COMPLETION MATCH NATURAL REALIZATIONS, OR ARE THERE OTHER POSSIBLE UV COMPLETIONS?

A. Goal

The goal of this step is to determine whether the minimal Class II UV completion, which we have shown to satisfy all the necessary physical conditions for TECT, is the only viable UV completion. Specifically, we want to address two key questions:

1. **Is this minimal Class II UV completion the one realized in nature?**
2. **Are there other UV completions that could lead to the same physical results, specifically the same shell-relevant tensor coefficients $A_{\text{eff}}, B_{\text{eff}}, C_{\text{eff}}$?**

By answering these questions, we can better understand whether the minimal Class II completion is truly unique or if there are alternative UV completions that could also work.

B. Physical Interpretation of the Minimal Class II Choice

The minimal Class II choice we derived leads to the following:

1. **Renormalization of Weyl coefficients:** The corrections $\delta A, \delta B, \delta C$ to the effective coefficients $A_{\text{eff}}, B_{\text{eff}}, C_{\text{eff}}$ are directly linked to the heavy auxiliary mediator field and lead to a renormalization of the Weyl coefficients.
2. **Weyl nondegeneracy:** The corrected Weyl criterion

$$A_{\text{eff}} \neq 0, \quad (B_{\text{eff}}, C_{\text{eff}}) \neq (0, 0)$$

implies that Weyl nondegeneracy is either stabilized or enhanced.

3. **Positive remote gap:** The correction to the Weyl determinant enhances the remote spectral gap Δ , making it positive, as required by the theory.
4. **Mass protection:** The minimal Class II completion maintains mass protection, ensuring no dangerous constant intervalley mass terms appear in the theory.

These results suggest that the minimal Class II choice is **physically consistent** and satisfies all the necessary conditions for a stable and viable microscopic realization of TECT.

C. Are There Other Possible UV Completions?

Given that the minimal Class II completion satisfies all the necessary conditions, the next question is whether there are other UV completions that could lead to the same physical results, i.e., the same shell-relevant tensor coefficients $A_{\text{eff}}, B_{\text{eff}}, C_{\text{eff}}$.

a. UV completion uniqueness. To determine whether other UV completions are possible, we need to consider the concept of **shell-equivalence**. Two UV completions are said to be shell-equivalent if, after integrating out heavy modes and projecting to the shell-relevant low-energy subspace, they yield the same effective coefficients.

Given the current results, the minimal Class II completion is **not the only possible UV completion**, as other UV structures could potentially lead to the same shell-relevant tensor coefficients. However, the minimal Class II choice is **the simplest realization** that leads to the correct physical behavior.

b. Alternative UV completions. Other possible UV completions could include:

- **Class I:** Direct shell-projector derivative couplings.
- **Class III:** Heavy connection or gauge-like mediators.
- **Other UV completions** that respect the symmetries of TECT and lead to the same shell-relevant tensor family.

However, **Class II remains the most natural and minimal** choice in terms of the number of parameters and the simplicity of the construction. Other UV completions may exist, but they are likely to be more complex or involve more parameters.

D. What Does This Mean for the Full TECT Framework?

For the full TECT framework, the minimal Class II UV completion provides a **simple and physically viable realization** of the theory. The corrections to the Weyl coefficients and the remote gap, as well as the preservation of mass protection, suggest that the minimal Class II choice leads to a physically stable and consistent theory.

Given the simplicity of the Class II completion and its ability to satisfy all the necessary conditions, it is likely to be **the primary candidate** for the correct microscopic realization of TECT.

E. Conclusion: Unique or Not?

The minimal Class II completion is not the **only** possible UV completion that could lead to the same shell-relevant tensor family. However, it is the **simplest** and most natural realization, and it satisfies all the necessary physical conditions for TECT. Other UV completions may exist, but they are expected to be more complex and are unlikely to lead to different physical behavior than the minimal Class II choice.

Thus, the minimal Class II UV completion is **the most likely candidate** for the correct microscopic realization of TECT.

The minimal Class II choice provides the simplest and most natural microscopic realization of TECT.

The next step is to further explore alternative UV completions and determine whether they lead to the same physical behavior as the minimal Class II completion.

XXX. COMPATIBILITY ANALYSIS OF THE CANONICAL MINIMAL CLASS II CHOICE WITH THE FULL DIRAC-CLOSURE CONDITIONS

A. Purpose

We now analyze the canonical minimal Class II representative not merely at the level of the Weyl coefficients, but against the full Dirac-closure conditions already isolated in the attached TECT manuscript.

At the theorem level, the shell part and the dynamical stability part are logically distinct:

1. the shell-relevant Weyl data are controlled by the three coefficients

$$A, \quad B, \quad C,$$

with

$$\det M = A(B^2 + C^2);$$

2. the full stable massless Dirac cone further requires:

$$V(0) = 0, \quad \Delta > 0, \quad \text{mass protection.}$$

Therefore the correct question is:

Does the canonical minimal Class II choice preserve the shell-relevant structure and remain compatible with the remaining Dirac-closure conditions?
(295)

B. Canonical minimal Class II representative

We fix the canonical shell-equivalent Class II representative by

$$m^a(P_0) = m \delta^{a3}, \quad m \neq 0, \quad (296)$$

and

$$\mathcal{C}_{ij}^{33} = c_{\parallel} \hat{n}_i \hat{n}_j, \quad (297)$$

$$\mathcal{C}_{ij}^{31} = c_I \hat{n}_i e_{1j} + c_J \hat{n}_i e_{2j}, \quad (298)$$

$$\mathcal{C}_{ij}^{32} = -c_J \hat{n}_i e_{1j} + c_I \hat{n}_i e_{2j}, \quad (299)$$

with all remaining components vanishing.

Define

$$\mu := \frac{\lambda g_X}{M_X^2} m. \quad (300)$$

Then the induced coefficient shifts are

$$\delta A = -\mu c_{\parallel}, \quad \delta B = -\mu c_I, \quad \delta C = -\mu c_J. \quad (301)$$

Hence

$$A_{\text{eff}} = A_0 - \mu c_{\parallel}, \quad B_{\text{eff}} = B_0 - \mu c_I, \quad C_{\text{eff}} = C_0 - \mu c_J. \quad (302)$$

The corresponding shell-relevant determinant is

$$\det M_{\text{eff}} = (\kappa_* |G_*|)^3 A_{\text{eff}} (B_{\text{eff}}^2 + C_{\text{eff}}^2). \quad (303)$$

C. I. Shell-level Weyl compatibility

This part is fully closed.

[Weyl compatibility of the canonical Class II representative] The canonical minimal Class II representative preserves the exact shell-relevant axial tensor family. Therefore the Weyl criterion remains

$$A_{\text{eff}} \neq 0, \quad (B_{\text{eff}}, C_{\text{eff}}) \neq (0, 0), \quad (304)$$

with

$$A_{\text{eff}}, B_{\text{eff}}, C_{\text{eff}}$$

given by (302).

By construction, the induced Class II tensor lies entirely in the same minimal axial family as the already-closed shell-relevant ansatz. Therefore it only renormalizes the three coefficients and leaves the form of the Weyl criterion unchanged.

Thus the canonical Class II choice is fully compatible with the shell part of the Weyl/Dirac theorem.

D. II. Opposite-valley pairing compatibility

The attached manuscript isolates opposite-valley pairing as a separate condition after the local Weyl data have been obtained. In the minimal shell-relevant theory, opposite-valley pairing is preserved when the coefficients at $+G_*$ and $-G_*$ match.

The canonical Class II choice is built from the shell-adapted axial objects

$$\hat{n}_i \hat{n}_j, \quad \hat{n}_i e_{1j}, \quad \hat{n}_i e_{2j},$$

in exactly the same way as the baseline shell-relevant tensor family. Therefore its behavior under opposite-valley exchange is controlled entirely by the parity of the background vector $m^a(P_0)$ and the mediator kernel $\mathcal{C}_{ij}^{ab}$.

[Sufficient condition for opposite-valley compatibility] If the canonical Class II data are chosen to be even under the matched opposite-valley transformation, then

$$\delta A^- = \delta A^+, \quad \delta B^- = \delta B^+, \quad \delta C^- = \delta C^+, \quad (305)$$

and therefore

$$A_{\text{eff}}^- = A_{\text{eff}}^+, \quad B_{\text{eff}}^- = B_{\text{eff}}^+, \quad C_{\text{eff}}^- = C_{\text{eff}}^+. \quad (306)$$

Hence the Class II correction preserves opposite-valley pairing.

The induced coefficients are linear in the background vector and the mediator kernel. If both transform evenly under the matched opposite-valley map, then the induced shell-relevant projections coincide at the two valleys. Since the baseline coefficients already match in the minimal shell construction, the renormalized coefficients also match.

So opposite-valley pairing is not automatic from the mere existence of Class II, but it is preserved under the natural evenness condition.

E. III. Compatibility with $V(0) = 0$

This point requires honesty.

The attached manuscript proves that the cleanest route to $V(0) = 0$ is a little-group mismatch between the enriched shell block and the remote sector at the crossing point, so that constant mixing is symmetry-forbidden. In that case one gets

$$V(0) = 0, \quad V(p) = O(|p|),$$

which is exactly the hypothesis needed in the Spectral Isolation Theorem.

The Class II correction does *not* by itself prove this. It only renormalizes the enriched shell tensor coefficients.

[Honest status of $V(0) = 0$ under Class II] The canonical minimal Class II choice is compatible with $V(0) = 0$, but does not by itself prove it. A sufficient condition remains:

$$\text{Hom}_{H_*}(H_R, H_D) = 0, \quad (307)$$

i.e. little-group mismatch between the enriched shell block and the remote sector at $p = 0$.

The Class II construction modifies the shell-relevant tensor data A, B, C , but does not itself classify the representation content of the full remote sector. Therefore the theorem-level sufficient criterion for $V(0) = 0$ remains the same as in the attached manuscript: distinct irreducible sectors imply no constant inter-sector mixing.

Thus:

Class II is compatible with $V(0) = 0$, but $V(0) = 0$ still has to be checked independently by little-group mismatch.

(308)

F. IV. Compatibility with the remote gap $\Delta > 0$

The attached manuscript isolates the remote-gap condition as an independent dynamical input:

$$\text{spec}(H_R(p)) \cap (-\Delta, \Delta) = \emptyset.$$

Again, the Class II choice does not by itself prove this. What it does do is preserve the shell-relevant structure of the enriched block. If the mediator is heavy and the induced correction is perturbative, then it should not close an already-existing remote gap. But this is a conditional stability statement, not an automatic theorem.

[Conditional gap compatibility] Suppose the unperturbed microscopic realization already has a remote gap

$$\Delta_0 > 0,$$

and suppose the Class II correction is perturbative in operator norm:

$$\|\delta H_{\text{Class II}}\| < \frac{\Delta_0}{2}. \quad (309)$$

Then the corrected model still has a positive remote gap

$$\Delta \geq \Delta_0 - \|\delta H_{\text{Class II}}\| > 0.$$

This is the standard stability of a gapped self-adjoint operator under a sufficiently small bounded perturbation. Therefore:

Class II is gap-compatible if it is a sufficiently small perturbation of an already-gapped microscopic realization. (310)

But the actual existence of $\Delta > 0$ is still a separate microscopic check, exactly as the attached manuscript states.

G. V. Compatibility with mass protection

The attached manuscript also isolates mass protection as an independent condition: a residual symmetry must forbid a constant Dirac mass term on the isolated enriched shell block.

The Class II canonical representative is built to renormalize the derivative shell tensor, not to insert a constant 4×4 mass operator directly. So it is structurally compatible with mass protection. But, again, this is not the same as proving mass protection.

[Honest mass-protection status] The canonical minimal Class II correction does not automatically generate a constant Dirac mass at the shell-relevant level, because it is constructed as a gradient-linear enriched correction. Therefore it is compatible with mass protection. However, whether the full corrected microscopic theory forbids all constant mass operators still depends on the residual symmetry algebra of the enriched shell block.

The induced correction changes A, B, C , i.e. the coefficients of the first-order shell operator. It does not by construction insert an independent constant shell-level Dirac mass term. But the absence of all symmetry-allowed constant masses is a stronger statement about the full low-energy symmetry algebra, not about the Class II correction alone.

Hence:

Class II is mass-protection compatible, but mass protection itself still has to be checked from the residual symmetry. (311)

H. VI. Final combined status

We can now state the exact logical status.

[Compatibility theorem for the canonical minimal Class II choice] Consider the canonical minimal Class II representative

$$m^a(P_0) = m\delta^{a3}, \quad \mathcal{C}_{ij}^{33} = c_{\parallel}\hat{n}_i\hat{n}_j, \quad \mathcal{C}_{ij}^{31} = c_I\hat{n}_ie_{1j} + c_J\hat{n}_ie_{2j}, \quad \mathcal{C}_{ij}^{32} = -c_J\hat{n}_ie_{1j} + c_I\hat{n}_ie_{2j},$$

with effective parameter

$$\mu = \frac{\lambda g_X}{M_X^2} m.$$

Then:

1. it preserves the exact shell-relevant axial tensor family and therefore the form of the Weyl criterion;
2. it preserves opposite-valley pairing provided the induced data are even under the matched opposite-valley transformation;
3. it is compatible with $V(0) = 0$, but $V(0) = 0$ still requires the independent little-group mismatch condition;
4. it is compatible with a positive remote gap $\Delta > 0$, but $\Delta > 0$ still requires an independent microscopic gap check (or small-perturbation stability from an already-gapped realization);

5. it is compatible with mass protection, but mass protection still requires the independent residual-symmetry exclusion of constant Dirac masses.

Items (1) and (2) follow from the shell-adapted form of the induced tensor. Items (3)–(5) follow because the Class II correction only renormalizes the derivative shell tensor coefficients and does not replace the already-isolated dynamical/symmetry hypotheses of the Dirac Emergence Closure Theorem and the Spectral Isolation Theorem.

I. What is actually closed after this step

What is now closed:

1. a concrete canonical minimal Class II representative exists;
2. it yields explicit

$$A_{\text{eff}}, B_{\text{eff}}, C_{\text{eff}};$$

3. the shell-level Weyl criterion is closed in terms of those effective coefficients;
4. opposite-valley pairing remains compatible under the natural parity condition.

What is still not closed:

1. the actual little-group mismatch proof for $V(0) = 0$;
2. the actual remote-gap proof $\Delta > 0$;
3. the actual residual-symmetry proof forbidding constant Dirac mass terms.

So the correct honest statement is not that Class II solves everything, but that it closes the shell-relevant tensor part and remains compatible with the remaining theorem-level hypotheses.

XXXI. FINAL SUMMARY

The canonical minimal Class II representative

$$m^a(P_0) = m\delta^{a3}, \quad C_{ij}^{33} = c_{\parallel}\hat{n}_i\hat{n}_j, \quad C_{ij}^{31} = c_I\hat{n}_ie_{1j} + c_J\hat{n}_ie_{2j}, \quad C_{ij}^{32} = -c_J\hat{n}_ie_{1j} + c_I\hat{n}_ie_{2j}$$

produces explicit effective coefficients

$$A_{\text{eff}} = A_0 - \mu c_{\parallel}, \quad B_{\text{eff}} = B_0 - \mu c_I, \quad C_{\text{eff}} = C_0 - \mu c_J, \quad \mu = \frac{\lambda g_X}{M_X^2} m,$$

and therefore the corrected Weyl determinant remains

$$\det M = (\kappa_*|G_*|)^3 A_{\text{eff}} (B_{\text{eff}}^2 + C_{\text{eff}}^2).$$

Thus the shell-level Weyl criterion is fully closed:

$$A_{\text{eff}} \neq 0, \quad (B_{\text{eff}}, C_{\text{eff}}) \neq (0, 0).$$

The Class II choice is also compatible with opposite-valley pairing, provided the induced data are even under the matched valley exchange.

However, the remaining full Dirac-closure conditions are not automatic consequences of Class II:

$$V(0) = 0$$

still requires little-group mismatch of the enriched shell block and the remote sector,

$$\Delta > 0$$

still requires an independent remote-gap check (or perturbative stability from an already-gapped realization),
 and mass protection still requires the residual symmetry to forbid all constant Dirac mass terms.
 Therefore the exact current status is:

Class II closes the shell-relevant tensor/Weyl part and is compatible with the remaining Dirac-closure hypotheses,

but

it does not by itself prove $V(0) = 0$, $\Delta > 0$, or mass protection.